International Journal of Heat and Mass Transfer 196 (2022) 123270

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Steady-state pool boiling heat transfer experimental studies of horizontally-placed tubes

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ARTICLE INFO

Article history:
Received 8 June 2022
Revised 19 July 2022
Accepted 21 July 2022
Available online 30 July 2022

Keywords:
Pool boiling heat transfer
Critical heat flux
Tubing wall thickness
Tubing materials
Accident tolerant fuel cladding

ABSTRACT

In this study, the pool boiling heat transfer experimental results are obtained from horizontally-placed tubes for various materials including FeCrAl alloys, Zircaloys, Inconels and Stainless Steels. The experimental results show that FeCrAl-C26M has higher pool boiling critical heat flux (CHF) than Zircaloys while the experimental CHF values of other two FeCrAl variants, i.e., B126Y and B136Y, are lower than that of Zircaloys. Given that FeCrAl-C26M and FeCrAl-C36M are reported to have higher pool boiling CHF than the traditional claddings, those experimental results speak to that the CHF priority of FeCrAl alloys depends on the variant series of tube materials. The micro-scale surface characterization analyses clearly illustrate that three FeCrAl variants are less susceptible to surface temperature overshooting oxidization than three Zircaloy variants. In comparison with FeCrAl-C26M, Inconel 600 has the closest pool boiling CHF among four Stainless Steels and three Inconels. Experimental results of this study support that both surface hydrodynamics and solid thermal-physical properties can have influential impacts on pool boiling CHF. It is also confirmed that the sandblasted and oxidized samples can render higher CHF values than their as-received samples. Moreover, the effects of tubing wall thickness on pool boiling CHF are experimentally investigated in this study. It is found that the corresponding experimental results show that thicker tubes result in lower pool boiling CHF.

1. Introduction

Boiling heat transfer is one of common heat removal strategies in thermal management systems such as the microelectronics coolers [1], industrial boilers [2], and light water reactors (LWRs) [3]. The thermal safety assessment of boiling heat transfer systems is determined by cladding surface Critical Heat Flux (CHF). There are several influential factors physically contributing to the CHF occurrence including surface morphological characteristics [4], heater dimension [5] and orientation [6], system conditions, and thermophysical properties of liquid/solid. However, the effects of heater thermophysical properties on boiling heat transfer still remain ambiguous while are worthy of systematic investigations. The use of optimal engineering materials can help improve the thermal efficiencies of boiling heat transfer systems without the help of micro/nano surface structures [2,7] or nanofluids [8].

Owing to the rapid advancing of high-entropy alloys (HEAs), the enlarging bank of thermal engineering materials enables versatile considerations to support the optimal design of thermal systems. For example, besides SiC/SiC_f, FeCrAl alloys, and Cr-coated Zircaloys, HEAs also seem promising candidates to the Accident Tolerant Fuel (ATF) claddings of LWRs [9]. The physical understanding to the mechanistic roles of thermophysical properties in boiling heat transfer are needed to support the future development of CHF-oriented engineering materials. This becomes more and more compelling since the advent of ATF concepts and the ATF-loaded core deployments. Several pool boiling experimental investigations have been performed across a variety of materials to analyze the contributions of thermophysical properties to significant CHF variations [10–14]. Tachibana et al. [11] pointed that the mechanistic aspects of hydrodynamics cannot rationalize pool boiling CHF variations between various materials with different wall thicknesses alone. The thermal activity (the product of wall thickness and thermal effusivity) was proposed to correlate the effects of thermophysical properties on pool boiling CHF [12,15]. Raghupathi and Kandlikar [13] thought that the square root of volumetric heat capacity was a better physical indicator than the thermal activity to predict the impacts of heater on pool boiling CHF. However, based on pool boiling CHF experiments of several ATF and traditional cladding materials, Yeom et al. [14] concluded that the effects of surface morphologies on pool boiling CHF are more dominant than that of thermal activity. Insofar, the effects of

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https://doi.org/10.1016/j.ijheatmasstransfer.2022.123270
0017-9310/

Nomenclature

Acronyms

ATF	accident nuclear fuel
CHF	critical heat flux (kW/m^2)
DC	direct current
FeCrAl	iron-chromium-aluminum
HL	heated length (mm)
HTC	heat transfer coefficient (kW/(m^2 K))
HEA	high entropy alloy
LWR	light water reactor
NBSS	nucleate boiling surface stability
NB-HTC	nucleate boiling HTC (kW/(m^2 K))
NC-HTC	natural circulation HTC (kW/(m^2 K))
ONB	onset of nucleate boiling (kW/m^2)
OD	outer diameter (mm)
PV	maximum peak-to-valley height (μm)
SEM	scanning electron spectroscopy
SS	stainless steel
WT	wall thickness (mm)

Symbols

α_s	thermal diffusivity (m^2/s)
θ_a	advancing contact angle (°)
Ra	averaged surface roughness (μm)
Fo*	boiling Fourier number (Dimensionless)
t_d	bubble departure time (s)
T	bulk temperature (°C)
R_e	electrical resistance (ohm)
T_s	interfacial surface temperature (°C)
ρ	mass density (kg/m^3)
V	power supply voltage (V)
r	radial distance of tubing (mm)
θ_r	receding contact angle (°)
c_p	specific heat capacity (J/(kg K))
θ_s	static contact angle (°)
q''	surface heat flux (kW/m^2)
ϕ	surface orientation angle (°)
k	thermal conductivity (W/(m K))
D_t	thermally equivalent diameter (mm)
Δ	uncertainty
U	voltage across the test section (V)

thermophysical properties on CHF still remain ambiguous and their mechanistic mechanisms behind CHF occurrence are controversial.

Besides the thermophysical properties, the geometrical dimensions and configurations also have influential impacts on thermal performances of pool boiling. For example, the pool boiling CHF of wire heater is significantly dependent of the wire diameter [16–19], and such this dependence is attributed to the hydrodynamics instabilities. This was also observed in the pool boiling CHF experiments using square plates [5] and circular disks [20]. Using the cylindrical vessel as the boiling chamber, the effects of confined space on pool boiling CHF were studied in the experimental investigations [20,21]. Both experimental studies gave a clear observation that increasing the vessel size can help enhance pool boiling CHF. It could be anticipated that the further increasing of vessel size beyond a certain point might not further enhance pool boiling CHF because the bubble behaviors adjacent to the heater-influenced region are not subjected to the confined space any more. The effects of the confined space and the heater size on pool boiling CHF matter a lot in the design of mini boiler systems such as the micro-chip cooling devices [22] and industrial small-scale boilers. It was postulated that the physical mechanisms behind the effects of heater shape, size and orientations are contributed by

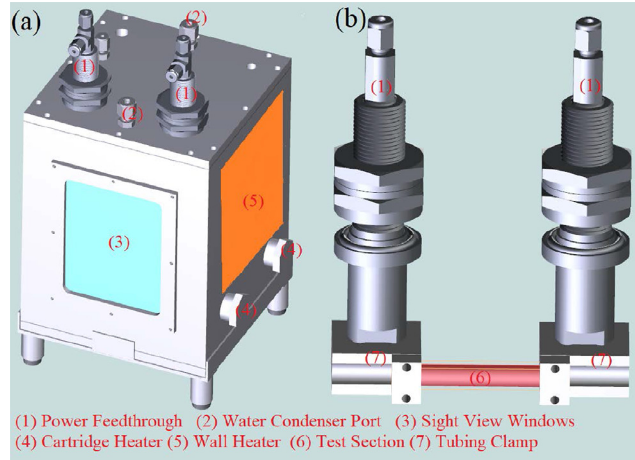


Fig. 1. Pool boiling experimental facility: (a) boiling chamber and (b) test section.

the interactive behaviors of bubble crowds that are usually demystified by hydrodynamics instability and nucleation site density theories [23]. However, the hydrodynamics-related theories cannot support the mechanistic rationalization to the effect of heater wall thickness on pool boiling thermal performances. For the plate-type heater, the parametric trend of pool boiling CHF shows an asymptotically increasing behavior with respect to the wall thickness [15]. But this experimental observation could not be applied to the cylindrical-type heater such as wire and tubing heaters.

In the present engineering material bank, there are many families of alloys for various industrial applications. For instance, Zircaloys and FeCrAl alloys are claddings of nuclear fuel pellets for the present and near-future concepts to prevent the release of radioactive elements into the water coolant. Inconel alloys are favorable candidates to chemically corrosive and high-temperature applications such as jet-engine, chemical reactors, and heat exchangers. Stainless steels are common engineering materials of culinary appliances, aerospace, and thermal power. Up to now, there were limited systematical investigations of pool boiling CHF experiments that were performed to study thermal efficiencies of those aforementioned materials. This study provides the comparative analyses for pool boiling performances differences of "with-family" and "between-family". Besides this, the effect of the tubing wall thickness on pool boiling CHF is experimentally studied using the exact same tubing materials. The rest of this paper is organized as follows, Section 2 describes the experimental methodologies, facilities and apparatus involved in the pool boiling CHF experiments, Section 3 presents the differences of surface morphological characteristics including micrographs and wettability before and after the CHF occurrences, Section 4 elaborates the experimental results analyses and discusses their potential implications, and Section 5 lists the key findings and some concluding remarks.

2. Experimental apparatus and methodologies

2.1. Pool boiling facility and test section

The pool boiling experimental facility is shown in Fig. 1 and the facility is a $180 \times 150 \times 230$ mm^3 stainless steel boiling chamber. In the pool boiling experiments of various alloys, the tested specimens of cladding materials are placed horizontally and submerged in the deionized water for CHF experiments. As shown in Fig. 1, the pool boiling facility is composed of several key parts: (1) two power feedthroughs are mounted on the chamber lid to apply direct-current programmable power to the test section, (2)

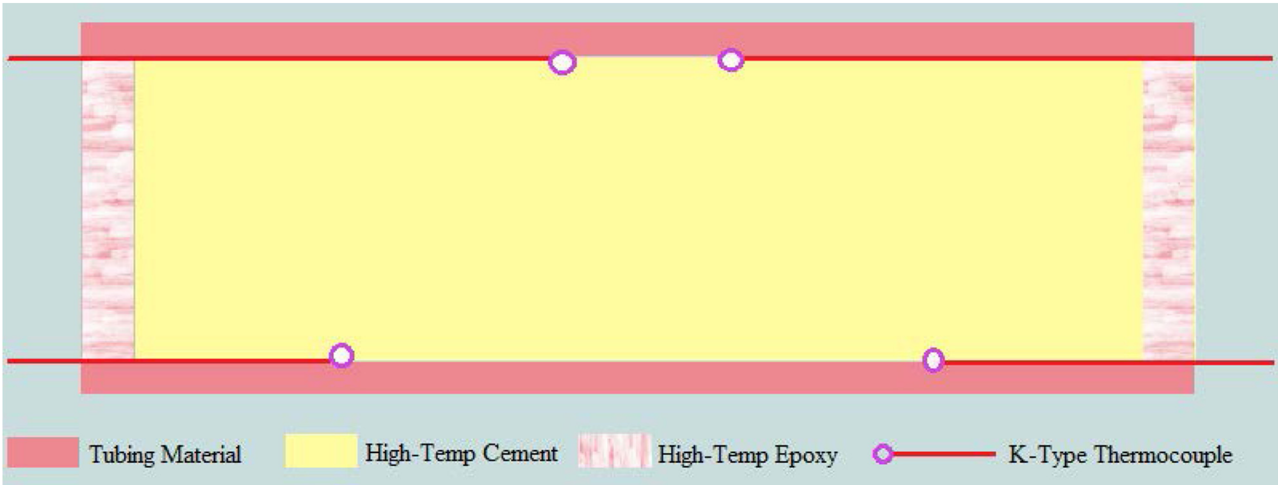


Fig. 2. Test specimen configuration.

a copper tube of surface condenser is fitted upon the chamber lid to cool the vapor steam back to the liquid state of water, (3) two sight view windows are mounted on the front and rear sides of chamber wall to capture bubble behaviors on the cladding materials by a high-speed camera, (4) two 400-Watt cartridge heaters are inserted in the pool boiling chamber to heat up the deionized water up to its saturation temperature, and (5) two 200-Watt wall heaters are pasted on the left and right sides of chamber wall to balance the thermal loss to the ambient environment.

In this pool boiling CHF experimental study, the working fluid is the deionized water. And the water level is 50 mm above the test section and it is maintained by a water-level sensor and a hot water tank to compensate the vapor loss to the ambient environment. The pool boiling facility is open to the ambient environment. In Albuquerque, the atmospheric pressure is ~84 kPa and the saturation temperature of water is 94.5°C. To minimize the disturbance of dissolved gas to the experimental results, this study adopts a degassing strategy that the saturation temperature of temperature is kept by two wall heaters working at least until there are no bubbles generated on the test section or chamber walls. Such this status is then kept for 45 min before performing experiments. The bulk temperature of the water pool is measured by a K-type thermocouple having a 1-mm sheathed diameter inserted through the chamber lid.

The test section is horizontally tightened between the copper clamps using four 6–32 stainless steel screws. The test section is a 3-inch-long tubing with the effective heated length of 2 inch. The 0.5-inch-long residuals are tightly fasted on the cylindrical grooves of power feedthroughs using the copper clamps. A voltage sensor is connected to two terminals of copper clamps to measure the voltage drop across the test section. As shown in Fig. 2, the test section is made of a metallic tubing, four K-type thermocouples, a high-temperature cement, and a high-temperature Alumilite's epoxy. The sample configuration of the test sections is prepared as follows: (1) the inner surface of cladding tube is rinsed respectively by acetone, alcohol and deionized water, (2) four K-type thermocouples are attached uniformly upon the inner surface of effective heated part based on adhesive silica pads, (4) the high-temperature cement (purchased from DAP Products Inc.) is used to fill the hollow space of tubing and both ends of tubing are open to the dry and room-temperature environment for seven days for the complete solidification of cement, (5) both ends of tubing are sealed by the water-proof high-temperature epoxy (purchased from Alumilite Corporation), and (6) the outer surface of the test section is also rinsed by acetone, alcohol and deionized water. Voltage, current and temperature signals are recorded using a data acquisition system purchased from National Instruments.

Two thermocouples are placed upon the inner surface at the down side of test sample 0.9 inch away from the sample ends respectively to detect CHF occurrence. Two other thermocouples are placed at the up side of test sample 1.3 inch away from the sample ends respectively to track the irreversible vapor formation spreading from the downside to the upside. It should be noted that the cement filler has an isothermal distribution of temperature under the steady-state condition. The corresponding analytical proof is given in Appendix A.1. This speaks to that temperature of the cement filler is theoretically same everywhere and the cement and epoxy thermal conductivities are not needed in the result analyses.

The uncertainty of thermocouples is ±1 °C according to the user manual of OMEGA Engineering Inc. The uncertainties of voltage and current transducers are 2.65% and 8.47% respectively at their maximum allowable scales. The data sampling frequency is 50 Hz and it means that there are 50 data points of each signal channel per second extracted from the data acquisition system. According to the user manual of OMEGA Engineering Inc, the uncertainties of voltage and current transducers are 2.65% and 8.47% respectively at their maximum allowable scales. The steady-state signifier of test section is that two time-wise averaged temperature points have no difference beyond ±1 °C provided that a time-wise temperature point is averaged over 50 consecutive temperature points from the same signal channel. On the other hand, the absolute difference between two time-wise average temperature points is beyond 30 °C or any temperature signal channel reads the inner surface temperature beyond 400 °C. This implies that CHF occurs on the cladding surface, or the boiling regime transits to the film boiling. Once either CHF or film boiling is detected, the power supply will be automatically shut down.

2.2. Tubing materials of nuclear fuel claddings

In this study, five families of cladding materials are chosen to cross-compare their experimental pool boiling heat transfer coefficients and CHFs. Their brief information is tabulated in Table 1.

Zircaloys The zirconium-based amorphous alloys have favorable characteristics in neutronics advantage of low thermal neutron absorption, high hardness, ductility, and corrosion resistance including Zircaloy 2, Zircaloy 4, Zirlo and M5 [24]. These Zircaloys are employed as the fuel cladding materials in the current power fleet of LWRs. However, the biggest flaw of Zircaloys is that the zirco-

Table 1
Brief information of test specimens.

Alloys	Types	OD (mm)	ΔOD (mm)	WT (mm)	Outer surface	$\Delta V/V$	$\Delta q''/q''$
FeCrAl	C26M	9.4996	0.0254	0.3810	As-Received Post-CHF Oxidized	0.005	0.0203
	B126Y	9.4742	0.0254	0.3556	As-Received Post-CHF Oxidized	0.005	0.0203
	B136Y	9.4996	0.0254	0.3810	As-Received Post-CHF Oxidized	0.005	0.0203
Zircaloy	Zircaloy 4	9.525	0.0381	0.508	As-Received Sandblasted	0.015	0.0304
	Zirlo	9.4742	0.0254	0.6350	As-Received Post-CHF Oxidized	0.017	0.0342
	Zr 705	9.525	0.0254	0.508	As-Received Post-CHF Oxidized	0.02	0.0402
Inconel	Inconel 400	9.525	0.0127	0.508	Sandblasted	0.009	0.0202
		9.525	0.0127	0.889	Sandblasted	0.013	0.0262
	Inconel 600	9.525	0.0127	0.508	As-Received	0.007	0.0202
		9.525	0.0127	0.7112	As-Received	0.009	0.0202
		9.525	0.0127	0.889	As-Received Post-CHF Oxidized	0.013	0.0262
	Inconel 625	9.525	0.0127	0.508	As-Received	0.008	0.0202
Stainless Steel	SS 304	9.525	0.0127	0.254	As-Received Post-CHF Oxidized	0.002	0.0202
		9.525	0.0127	0.381	As-Received Post-CHF Oxidized	0.003	0.0202
		9.525	0.0127	0.508	As-Received Post-CHF Oxidized	0.004	0.0202
		9.525	0.0127	0.7112	As-Received	0.005	0.0202
		9.525	0.0127	0.889	As-Received	0.008	0.0202
	SS 316	9.525	0.0127	0.254	As-Received	0.003	0.0202
		9.525	0.0127	0.381	As-Received	0.005	0.0202
		9.525	0.0127	0.508	As-Received Post-CHF Oxidized	0.006	0.0202
		9.525	0.0127	0.7112	As-Received	0.009	0.0202
		9.525	0.0127	0.889	As-Received	0.011	0.0222
	SS 321	9.525	0.0127	0.508	As-Received Post-CHF Oxidized	0.007	0.0202
	SS 347	9.525	0.0127	0.508	As-Received Post-CHF Oxidized	0.005	0.0202
Titanium	Grade-2	9.525	0.0381	0.889	Sandblasted	0.025	0.0502
	Grade-9	9.525	0.0381	0.889	As-Received Post-CHF Oxidized	0.025	0.0502

Noting that some samples are sandblasted by the sand paper (grit number is 180) while others are oxidized by the surface temperature overshooting of pool boiling CHF occurrence.

nium can fiercely react with the hot water steam at the temperature beyond 1230 °C [24]. And oxidation of zirconium by water is accompanied by release of hydrogen gas and of exothermic heat [24]. Up to now, most of commercial LWRs are still on the use of UO_2-Zircaloy fuel-clad systems. The available experimental data of T-H Zircaloy characteristics are still not sufficient to support the comprehensive thermal safety assessment of Zircaloy claddings.

FeCrAl alloys As alternative candidates to replace zirconium-based alloys, these FeCrAl alloys have better potentials in resistance to high-temperature steam oxidation, material structural strengths, and the resistances to irradiation and chemical corrosion [25]. However, the FeCrAl alloys have higher absorption cross section of thermal neutrons. This makes the energy spectrum of neutrons hardening, resulting in unwanted outcomes including the reactivity suppression and the higher burnup [25]. The T-H characteristics of FeCrAl alloys are of importance to license the ATF-loaded LWRs in the near-term implementation of ATF.

Stainless Steels Austenitic stainless steels are widely used as in-core and surrounding structural materials in current commercial LWR systems. Moreover, the stainless steels are often used as the reference materials in most of T-H experiments. In the early stage of nuclear energy, the stainless steels were used as claddings to encase the UO_2 fuel pellets [26]. This study adopts the stainless steels as the references basis to compare with T-H characteristics of other cladding materials.

Nickel-based alloys Nickel alloys are used extensively in nuclear reactor applications because they have high strength and good corrosion properties. However, nickel has some unique nuclear properties (including relatively high thermal neutron absorption cross-sections) that limit its use in power reactor cores. The primary application of nickel alloys is to manufacture the steam generator of LWRs [27]. Besides the nuclear energy applications, nickel alloys are also used as the heating elements in the chemical and petrochemical industries. The T-H characteristics of nickel alloys can be transferred from the nuclear power to the chemical-related industries.

Besides four alloy families above, two titanium alloys are also experimentally investigated in this study for their pool boiling CHF. Because of their unique properties in corrosion-resistance and strength-to-weight ratio, they are widely applied in high temperature environments such as jet engines.

2.3. Contact angle goniometer, high-speed camera, surface characterizations

The surface roughness and contact angles are measured with a surface profilometer and contact angle Goniometer, respectively. The surface profilometer (Dektak 150, Fig. 3a) is a stylus type with 1 mg force to measure surface peaks and valleys. The average roughness, Ra, of each scan is calculated from the surface topology measurements using the Dektak software. The machine can measure samples up to 5 inch in diameter. The system has a 1-millimeter standard vertical range with up to 120,000 data points per scan. The machine is capable of measuring surface flatness, waviness, and radius of curvature as well as characterizing thin-film stress on wafers.

The contact angle goniometer is a device that can measure contact angle, surface tension, and surface energy. The facility is fitted with an automated dispensing system to control the droplet volume and ensure measurement repeatability. The automated tilting base is a fully controlled base to change the specimen tilting angle and measure both advance and receding contact angles, θ_a and θ_r, respectively. The machine is also equipped with a temperature-controlled hot chamber for conducting measurements at an elevated temperature up to 300°C (Fig. 3b). The measurement procedures of surface contact angles are practiced on the basis of the American society of testing materials standards [28,29]. A gray-scale high-speed optical camera is employed to record the live video of boiling dynamics behaviors on the cladding tube surface (the camera is purchased from the Fastec Motion Inc., Ts model). This helps demystify the CHF triggering mechanisms on the horizontally placed tube surface. In this experimental studies, the frame per second rate is fixed at 1000 Hz. The micro-scope analysis machines including the optical scan, scanning electron microscopy (SEM) scan are utilized to characterize the surface morphological features before and after CHF occurrence. This benefits to the un-

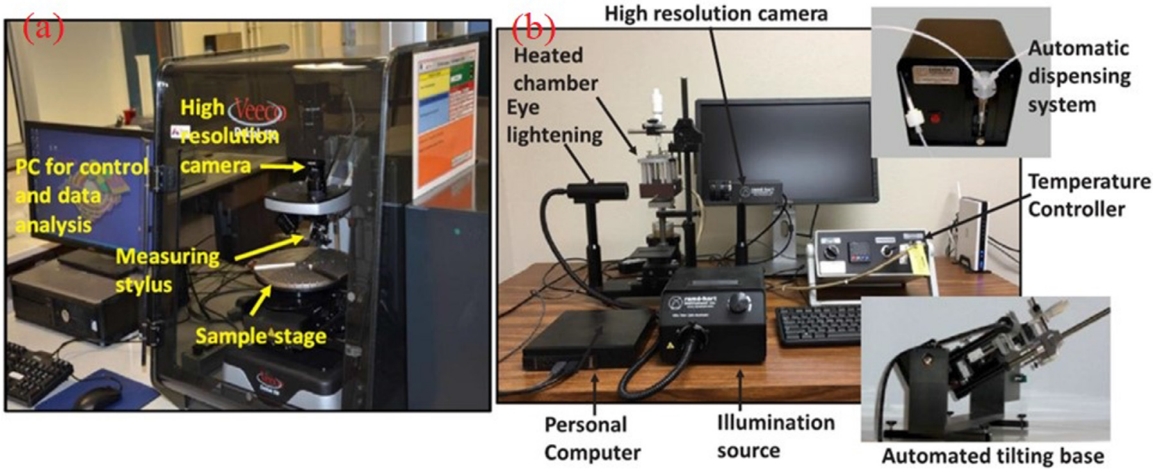


Fig. 3. Dektak 150 surface profilometer (left) and contact angle goniometer (right).

(a) Three FeCrAl alloys (b) Three Zircaloys

Fig. 4. Post-CHF samples of ATF and traditional claddings .

derstanding of how boiling heat transfer contributes to the surface oxidization/corrosion.

3. Cladding surface characterizations and bubble behaviors

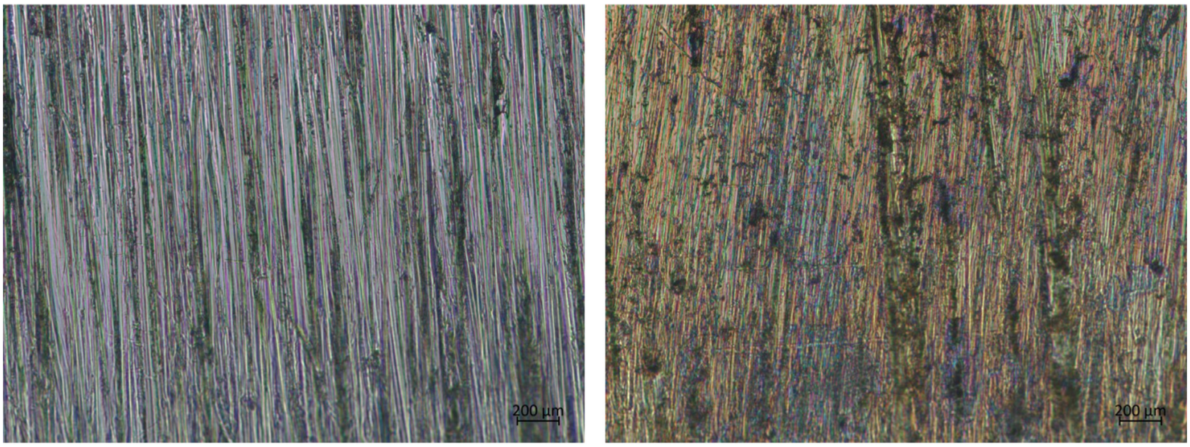
3.1. Micrographs of optical scans

As shown in Fig. 4(a), FeCrAl-C26M has a better corrosion resistance[1] than FeCrAl-B126Y and FeCrAl-B136Y to the pool boiling CHF occurrence from the visual appearance of post-CHF surfaces. The evident features on the cladding surfaces of B126Y and B136Y are a slim crack and a cluster of oxidized spots respectively.

In comparison with the post-CHF surfaces of three Zircaloys, FeCrAl alloys have better corrosion resistance to the CHF-occurrence. Because the cladding surfaces of three Zircaloys are totally covered by a layer of oxides. Figs. 5, C.24 and C.25 respectively show the micro-scale surface appearances before and after the pool boiling CHF experiments for three FeCrAl alloys, C26M, B126Y and B136Y. It is obvious that in comparison with B126Y and B136Y, there are less clusters of oxide deposits on the post-CHF surface of C26M. Besides, the average size of oxide deposits on the FeCrAl-C26M post-CHF surface is smaller than that of B126Y and B136Y. The slim crack on the post-CHF surfaces is located upon the downside of the horizontally placed FeCrAl-B126Y tube. However, the cluster of oxidized spots is observed in the downside of the horizontally placed FeCrAl-B136Y tube. This physically means that the surface oxidization corrosion on the downside is worse than on the upside. The post-CHF surface of FeCrAl-C26M shows the similar observations of oxidization corrosion. These optical micrographs of post-CHF surfaces imply that CHF firstly occurs on the downside of horizontally placed tube and then extends to the upside.

Figs. C.26 –C.28, present the microscale optical scans of Zircaloy 4, Zirlo and Zr 705 respectively before and after pool boiling CHF experiments. From the comparisons between the post-CHF surfaces of Zircaloy 4 and Zirlo, there is no appreciable difference. However, it is observed that there are some unknown crystal features on the post-CHF surface of Zr 705.

[1] Herein, the authors connote the corrosion resistance as the ability of cladding material to withstand the surface temperature overshooting and generate thinner oxide layer.



(a) As-Received (b) Post-CHF

Fig. 5. Optical scan of FeCrAl-C26M before and after boiling at the magnification of 20.

Fig. 6 compares the post-CHF surfaces between Zircaloys and FeCrAls and presents some typical surface morphological features resulting from the pool boiling CHF occurrence. From the oxide distributions on the post-CHF surfaces, FeCrAl-C26M has the best resistance ability to the oxidization corrosion at the surface temperature overshooting of CHF occurrence. Among three Zircaloys, the Optimized Zirlo has a better corrosion resistance to the surface oxidization.

Table 2
Oxide layer thicknesses of Zircaloys and FeCrAls obtained by SEM.

Claddings	Upside: $\phi = 0^{\circ}(\mu m)$	Downside: $\phi = 180^{\circ}(\mu m)$
FeCrAl-C26M	0.09 ± 0.03	0.27 ± 0.02
FeCrAl-B126Y	0.29 ± 0.04	0.94 ± 0.05
FeCrAl-B136Y	0.18 ± 0.06	0.81 ± 0.09
Zircaloy 4	6.13 ± 0.23	11.22 ± 0.37
Zirlo	4.18 ± 0.36	9.53 ± 0.62
Zr 705	6.85 ± 0.28	12.64 ± 0.52

3.2. SEM micrographs analyses

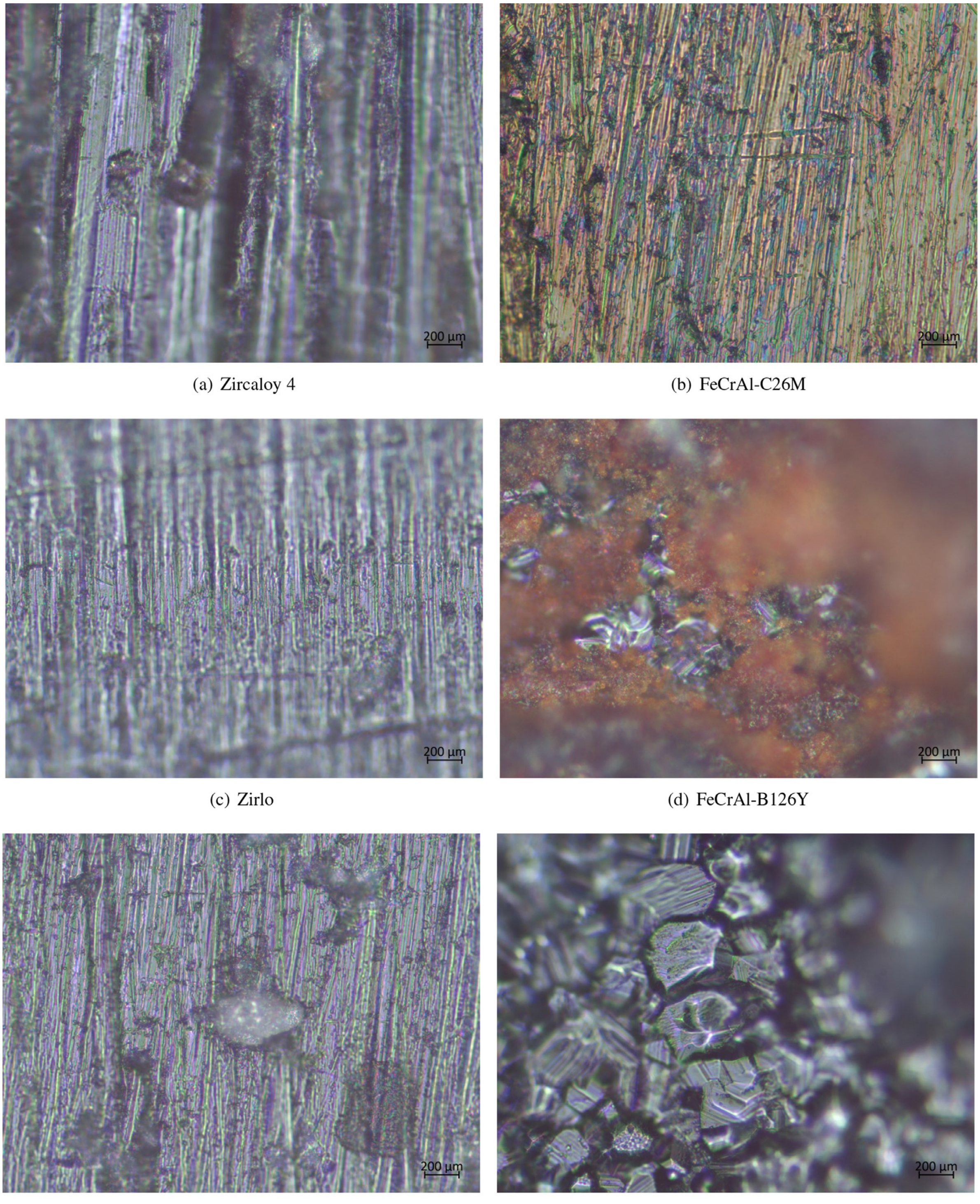
Figs. 7, C.29 and C.30 present SEM scan micro-graphs of as-received and post-CHF surfaces respectively for FeCrAl-C26M, FeCrAl-B126Y and FeCrAl-B136Y before and after the pool boiling CHF experiments. As show in three figures, the oxide layers of both FeCrAl-B126Y and FeCrAl-B136Y are stripped off from the post-CHF surfaces, which results in the further surface oxidation while the oxide layer of FeCrAl-C26M still remains intact to protect cladding surface from the high-temperature steam corrosion. On the other hand, the average size of oxidized features on the post-CHF surface of FeCrAl-C26M is smaller than that of FeCrAl-B126Y and of FeCrAl-B136Y. These micro-scale observations made to post-CHF surfaces of three FeCrAl alloys speak to that FeCrAl-C26M has the best corrosion resistance to the high-temperature water/steam mixture. Besides the comparisons between as-received and post-CHF surfaces, some unique features resulting from pool boiling CHF occurrence are also presented respectively for FeCrAl-C26M (See Fig. C.31), FeCrAl-B126Y (See Fig. C.32), and FeCrAl-B136Y (See Fig. C.33). By cross-comparing the post-CHF surfaces of three FeCrAl variants, it is convinced that FeCrAl-C26M is less susceptible to the corrosion impacts of CHF occurrence than FeCrAl-B126Y and FeCrAl-B136Y.

Figs. C.34 –C.36 show SEM scanning micrographs of as-received and post-CHF surfaces respectively for Zircaloy 4, Zirlo and Zr 705. Besides the oxide layer covered on the post-CHF surfaces, there are some unique crystal deposits presented on surfaces. The X-ray photoelectric spectroscopy analyses imply that the crystal deposits are zirconium hydrides (See Figs. C.35(b) and C.36(b)). It was widely accepted that FeCrAl alloys have better corrosion resistance ability to the high temperature steam/water mixture than Zircaloys [30]. The qualitative analyses by the micro-graphs of post-CHF surface SEM scanning can not provide the overall comparisons of corrosion resistance ability between 6 different nuclear cladding materials.

In this study, the post-CHF cladding samples are machined to produce the cross-section view for the SEM scanning measurement of oxide layer thickness. As shown in Fig. 8, there is a clear boundary between the oxide layer and base material. Given that the boiling behavior difference between the upside and downside of tested samples, the oxide layer thickness of each side is averaged by 18 measurement points respectively. For example, 6 SEM scan locations are picked up on the upside of post-CHF surface and measuring the oxide layer thickness is performed at 3 different points. Then the oxide layer thickness of the tested sample upside is averaged by 18 different measurement points.Provided in Table 2, the oxide layer thickness of upside is less than that of downside. This confirms that the oxidization corrosion of downside is worse than that of upside, possibly implying CHF may firstly occur on the downside of horizontally placed tube. Also, FeCrAl-C26M has the least oxide layer thickness among six nuclear fuel cladding materials, which gives a direct proof to that FeCrAl-C26M has the best corrosion resistance to the high-temperature surface oxidization. Among three Zircaloys, Zirlo has the thinnest oxide layer, which is in agreement with the results analyses of Fig. 6.

3.3. Surface wettability and roughness on cylindrical surfaces

The post-CHF surfaces of FeCrAl alloys become more hydrophilic than their as-received surfaces (See Figs. 9, C.37 and C.38). It is noteworthy that the pool boiling CHF occurrence on the cladding surface can help reduce the surface wettability difference between as-received samples after the boiling experiments. This surface wettability change is also observed in the post-CHF surfaces of Zircaloys (See Figs. C.39–C.41). However, the post-surfaces of six nuclear claddings are more hydrophilic than their counterparts of as-received samples. This results in that the CHF re-occurrence on the post-CHF samples needs more thermal energy to take place, further potentially increasing the safety margins of



(a) Zircaloy 4

(b) FeCrAl-C26M

(c) Zirlo

(d) FeCrAl-B126Y

(e) Zr 705

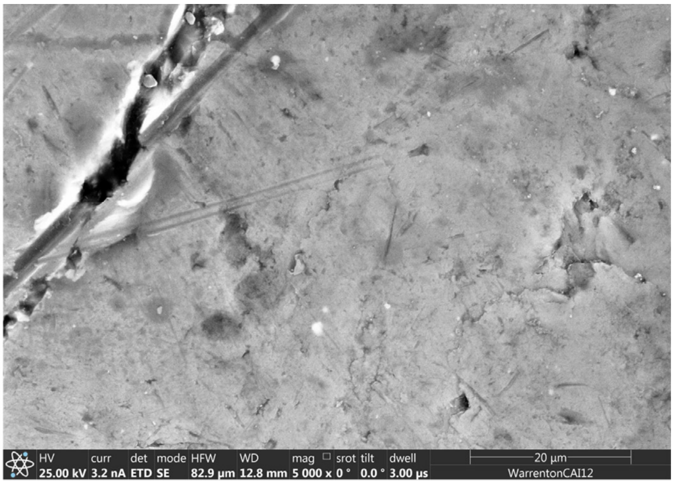
(f) FeCrAl-B136Y

Fig. 6. Optical scan of FeCrAl alloys and Zircaloys for typical micro-scale surface features at the magnification of 50 .

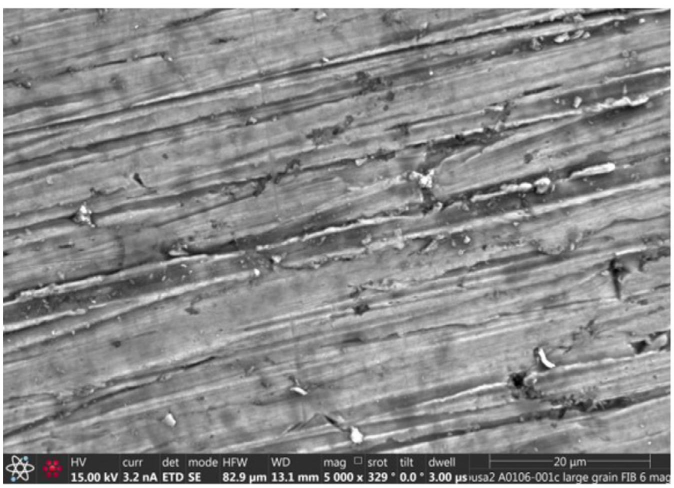
thermal management system. However, this superiority of pool boiling CHF enhancement could be gradually weaken by the increasing of liquid subcooling.

Although the as-received FeCrAl samples are less hydrophillic than the as-received Zircaloy samples, the post-CHF surfaces of FeCrAl samples are more hydrophillic than the post-CHF surfaces of Zircaloys. The surface wettability enhancement by oxide layers in FeCrAl alloys is greater than that of Zircaloys. This further confirms that besides surface morphological features, the chemical constitutions of cladding materials can affect the surface wettability significantly.

Based on Kandlikar's CHF model [31], the receding contact angle is the study of interest that explains the effect of surface wettability on CHF. The contact angle goniometer (Rame-Hart) can measure both advance and receding contact angles by changing the specimen tilting angles and using the automated tilting base. The static contact angle can be used as an indicator of surface wettability of oxidized samples in this study, and it is measured on the as



(a) As-Received



(b) Post-CHF

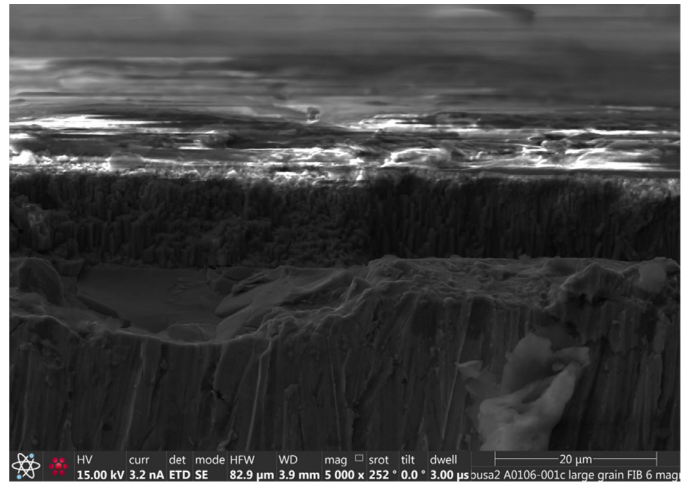
Fig. 7. SEM scan of FeCrAl-C26M before and after pool boiling CHF experiment.

received and post-CHF surfaces. The corresponding measurement procedures are in accordance with the standard of American Society of Testing Materials [28,29]. The ambient temperature of contact angle measurement is approximately close to the inlet temperature of water coolant, and the tested specimens are put in a transparent chamber where the ambient temperature is controlled by the contact angle goniometer. The surrounding gas medium of the transparent chamber is the dry air. To minimize the volume shrinkage of water droplet, five measurements are performed on each sample within 5 secs respectively. Table 3 provides the measured contact angles of tested cladding materials. As shown in Table 3, the receding angle is less than the static contact angle. The results of contact angle can be validated by the Tadmor model of surface wettability [32]. This model relates the static contact angle with the advancing and receding contact angles based on the equilibrium line energy as the follow three sets of equations,

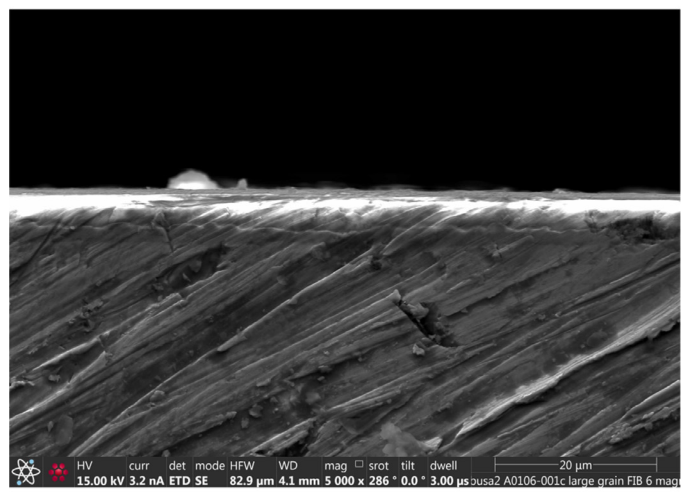
$$\theta_s = \arccos\left(\frac{\Gamma_a\theta_a + \Gamma_r\theta_r}{\Gamma_a + \Gamma_r}\right) \tag{1a}$$

$$\Gamma_a = \left(\frac{\sin^3(\theta_a)}{2 - 3\cos(\theta_a) + \cos^3(\theta_a)}\right)^{1/3} \tag{1b}$$

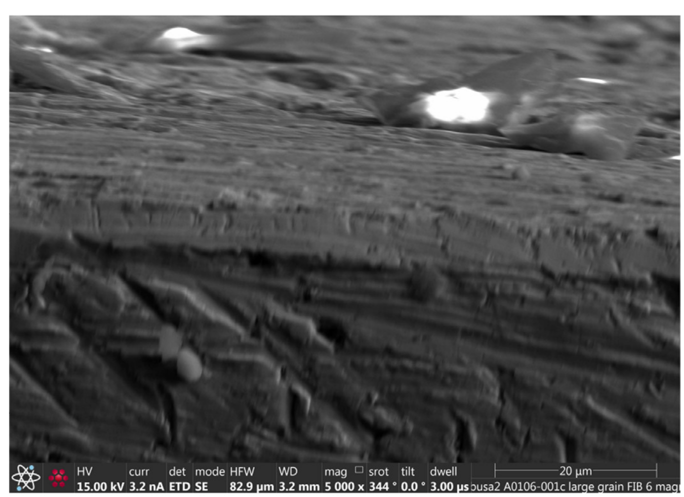
$$\Gamma_r = \left(\frac{\sin^3(\theta_r)}{2 - 3\cos(\theta_r) + \cos^3(\theta_r)}\right)^{1/3} \tag{1c}$$



(a) Zircaloy 4



(b) Zirlo



(c) Zr 705

Fig. 8. SEM scan for the cross-section views of oxide layer to three Zircaloys.

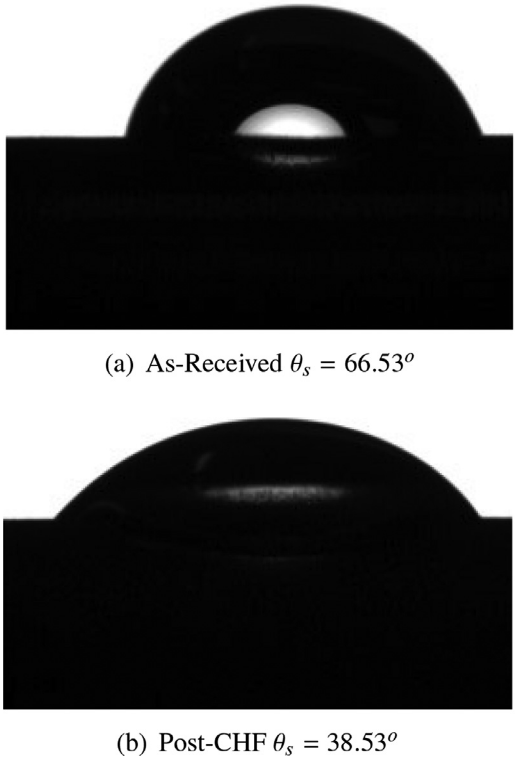


Fig. 9. Surface static contact angles of FeCrAl-C26M before and after pool boiling CHF experiment.

Table 3
Contact angles of tested samples.

Alloys	Types	Contact angles		
		θ_s (°)	θ_r (°)	θ_a (°)
FeCrAl	C26M	66.53	54.26	94.88
	B126Y	74.86	63.45	97.78
	B136Y	81.84	67.28	104.79
Zircaloys	Zircaloy 4	65.71	49.3	83.4
	Zirlo	63.87	48.34	84.83
	Zr 705	57.11	40.62	85.73
Inconel	400	67.92	55.7	98.49
	600	65.54	55.69	93.23
	625	62.14	50.16	88.23
Stainless Steel	304	70.77	63.36	87.16
	316	73.7	65.34	91.35
	321	77.62	68.56	92.09
	347	72.14	64.21	89.75
Titanium	Grade-2	54.12	40.36	72.11
	Grade-9	59.56	47.41	73.59

Noting that static angle θ_s, advancing angle θ_a, and receding angle θ_r, all the angles are measured upon the horizontally placed tubings.

The predicted static contact angle based on the Tadmor model is compared with the measured static contact angle for all the tested materials as shown in Fig. 10. The 10.6% model discrepancy possibly results from the measurement uncertainty and the curvature effect that is not incorporated in the Tadmor model. Because the Tadmor model is derived for contact angle relations on the plates instead of tubing.

Surface roughnesses of tested samples are obtained by profilometer to provide a cross-comparative analysis with thermal-physical properties in the pool boiling heat transfer (See Table 4). A 3-inch-long track line is randomly selected from the axial direction of test samples to perform surface roughness measurements. Although Ra varies from 1.55 μm to 6.84 μm between different as-received samples, the as-received samples could be still considered consistent in surface morphologies. Under this assumption, the difference gap of pool boiling between claddings could be reasonably attributed to solid thermal-physical properties. However, given that some samples are sandblasted by sandpapers with grit 180, those rougher surfaces can exert more influential impacts on pool boiling heat transfer than the as-received samples.

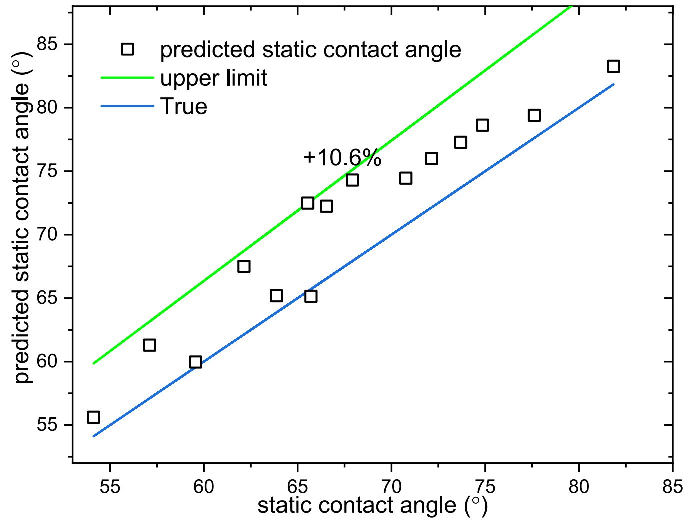


Fig. 10. Measured static contact angle against the predicted static contact angle using the Tadmor model.

Table 4
Surface roughnesses of tested samples.

Alloys	Types	Surface roughnesses (μm)		
		Ra	PV	SM
FeCrAl	C26M	2.12	16.93	24.26
	B126Y	4.86	23.45	27.79
	B136Y	6.84	27.28	30.97
Zircaloys	Zircaloy 4	5.11	24.39	33.4
		23.62§	56.23§	69.84§
	Zirlo	3.76	23.36	28.59
	Zr 705	4.15	20.42	35.37
Inconel	400	19.29§	55.72§	78.42§
	600	1.55	19.96	21.32
	625	2.64	21.01	24.38
Stainless Steel	304	4.77	21.33	26.16
	316	5.97	20.13	22.35
	321	5.62	18.56	26.09
	347	4.72	16.64	28.75
Titanium	Grade-2	25.21§	54.11§	72.36§
	Grade-9	6.55	27.14	33.75

Noting that Ra is the averaged surface roughness,PV is the maximum peak-to-valley height, SM is the mean spacing between peaks, § is the sandblasted samples by the sand papers with grit 180, otherwise samples are as-received from vendors. OD is the outer diameter and WT is the tubing wall thickness.

3.4. Bubble behaviors

In the pool boiling experiments of horizontally placed tubes, the surface orientation angle with respect to the motion direction of bubble varies from 0° (up-side) to 90° (front/rear-side) to 180° (down-side). The bubble behaviors difference on three sides result from the surface orientation angle (See Fig. 11).

Figs. 12, D.42 and D.43 show the bubble behaviors on the horizontally-placed claddings of three FeCrAl and three Zircaloys respectively at the heat flux of 51, 139 and 206 kW/m^2. As observed in Fig. 12, the bubble nucleation site density of three FeCrAl alloys is greater than that of three Zircaloys. It is noteworthy that the bubble diameter on the down-side is larger than those on the up-side. One possible phenomenological explanation behind this is

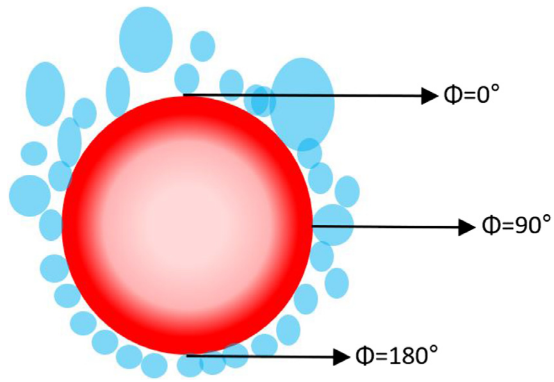


Fig. 11. Schematic of boiling behaviors on a horizontally placed tube.

that the cladding tube can impede the bubble departure from the down-side surface. In light of this bubble impedance by the down-side surface, the CHF may more likely occur on the down-side surface of cladding tube. As further increasing heat flux, this bubble impedance by the down-side surface becomes more and more evident. In comparison with the nucleate site density of FeCrAl-B126Y and FeCrAl-B136Y, the FeCrAl-C26M cladding surface has a less nucleate site density and generates larger bubbles. This can help delay the occurrence of pool boiling on the FeCrAl-C26M cladding surface and thus result in the higher pool boiling CHF than FeCrAl-B126Y and FeCrAl-B136Y.

Besides the cladding materials and surface wettability, the surface roughness can have influential impacts on the bubble dynamics behaviors. For the horizontally-placed tubes of Inconel alloys and stainless steels with exactly same outer diameter and wall thickness, the cladding surfaces of tested samples are blasted by the 800-grit sandpapers on the automatic machining machine to generate approximately same surface-roughness scales. As shown in Figs. D.44–D.46, the Inconel 600 and 625 generate less crowds of larger bubbles than SS 304, 316, 321, and 347 under three different surface heat flux. Given that all cladding surfaces are blasted by the 800-grit sandpapers, the one possible plausible rationale behind this difference of bubble behaviors could result from the surface activation energy that needs to activate nucleate sites in relation with the thermal-physical properties.

It is observed in Figs. 12 to D.46 that the bubble behaviors are roughly similar on the cladding surfaces of “within families” materials while show significantly different patterns on the “between families” materials. For example, the bubble dynamics behaviors show no appreciable difference between Inconel 600 and Inconel 625, between FeCrAl alloys, or between Zircaloys while it is directly evident that bubble dynamics patterns of stainless steels are significantly distinct from those of Inconel alloys, and of FeCrAl alloys. This could be attributed to the thermal-physical properties variations of the “between families” materials. On the other hand, the difference of surface morphological features between various families of alloys could rationalize the bubble dynamics behaviors from the perspective of nucleate site density theories. This assumption that thermal-physical properties may play a decisive role in the bubble dynamics behaviors will be left for the future experimental validations using the square plate cladding made of various solid materials. In this study, the horizontally-placed tubing can not support the quantitative and parametric analyses of bubble dynamics behaviors including the bubble departure diameter and frequency because of the curvature effect of cylindrical tubing.

Considering that FeCrAl alloys have a higher likelihood of post-CHF survival and a better resistance to the structural failure resulting from CHF occurrence, the bubble behaviors on the post-CHF surfaces of FeCrAl alloys are also of importance to the physical un-

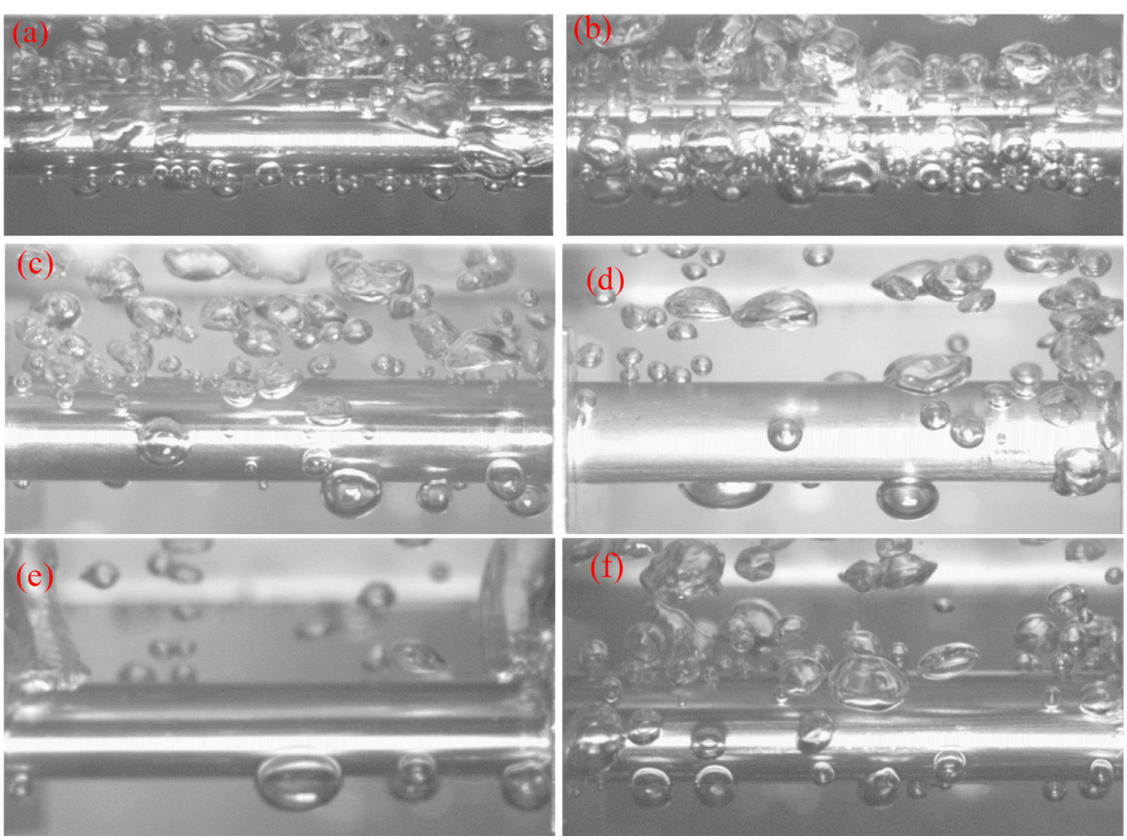


Fig. 12. Bubble behaviors of three Zircaloys and three FeCrAl alloys at the heat flux of 51 kW/m^2:(a) FeCrAl-B126Y, (b) FeCrAl-B136Y, (c), FeCrAl-C26M, (d) Zircaloy 4, (e) Zirlo, and (f) Zr 705.

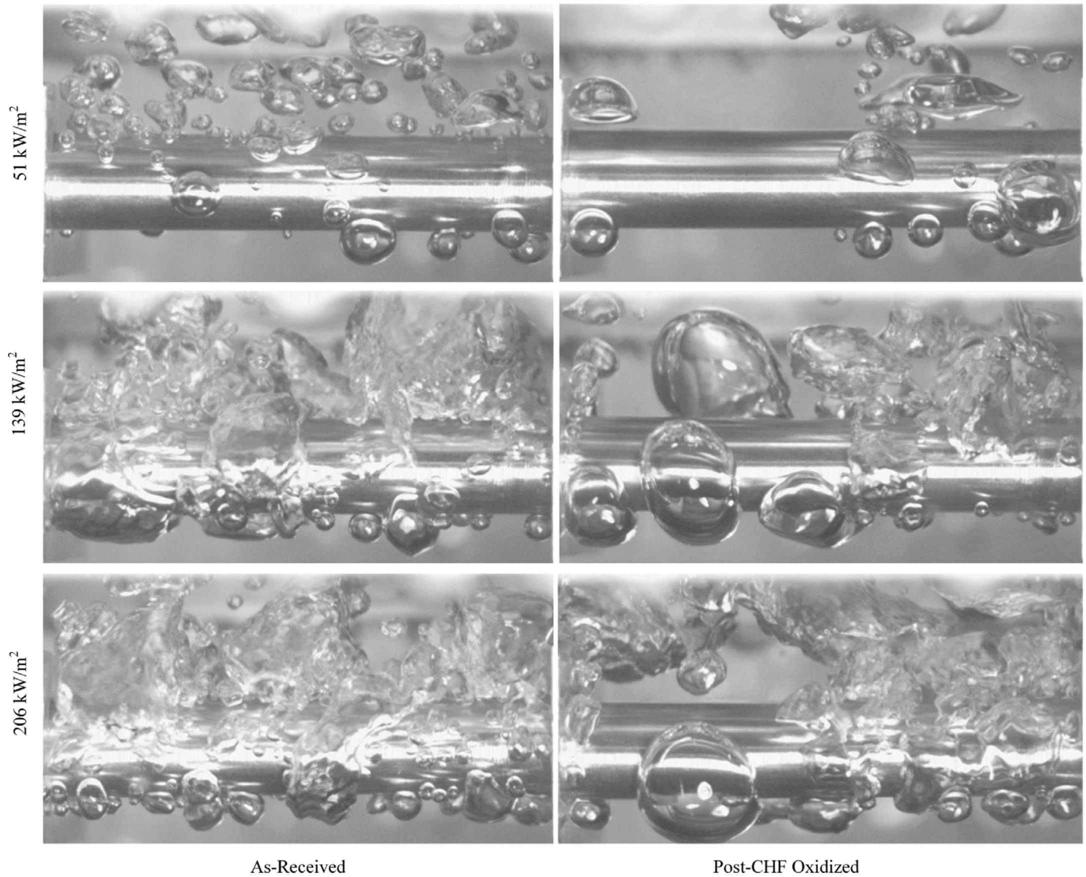


Fig. 13. Bubble behaviors of as-received and post-CHF FeCrAl-C26M .

derstanding to boiling heat transfer mechanisms that are subjected to the impacts of oxide layer. Figs. 13 and 14 demonstrate that the oxidized region of post-CHF samples generates larger bubbles but sparser bubble crowds. The introduction of oxide layer to the boiling heat transfer interface brings variations in two factors affecting CHF triggering mechanisms, surface morphological features including oxide clusters and micro-scale cracks (See Fig. C.33), and thermal-physical properties alteration cross the triple line of boiling as a result of oxide layer presence.

4. Results and discussions

4.1. Pool boiling experimental results

The pool boiling curves of three FeCrAl alloys are shown in Fig. 15(a) and their heat transfer coefficients are presented in Fig. 16. As observed, the natural circulation heat transfer coefficient (NC-HTC) is approximately same for three different FeCrAl alloys before the onset of nucleate boiling (ONB). However, beyond ONB, the nucleate boiling heat transfer coefficient of (NB-HTC) FeCrAl-B136Y is less than that of FeCrAl-C26M and FeCrAl-B126Y. But there is no appreciable difference of NB-HTC between FeCrAl-C26M and FeCrAl-B126Y. For pool boiling CHF of saturated water, FeCrAl-C26M shows better thermal priority over FeCrAl-B126Y, and FeCrAl B136Y.

The pool boiling curves and heat transfer coefficients of three Zircaloys are respectively presented in Figs. 15(b) and 17. Similar to the pool boiling cases of FeCrAl alloys, there is no appreciable difference of NC-HTC between three Zircaloys. However, beyond ONB, NB-HTC of Zirlo is greater than that of Zircaloy 4 and Zr 705. It is noteworthy that for Zirlo, the fully-developed nucleate boiling occurs because as shown in Fig. 17, the further increasing of heat flux can not enhance the pool boiling NB-HTC of Zirlo. Fig. 18 demonstrates that Zirlo has a comparable NC-HTC to that of FeCrAl-C26M while beyond ONB point, FeCrAl-C26M has a pool boiling higher NB-HTC than Zirlo. It is noteworthy that there is no characteristics of fully-developed nucleate boiling reflected in the HTCs of FeCrAl alloys. One possible explanation behind this is that the fully-developed nucleate boiling heat flux of FeCrAl alloys is suppressed by the hydrodynamics instability occurrence on the horizontally-placed tubing.

Tables 5–7 tabulate the CHF experimental results of horizontally-placed tubes respectively for nuclear claddings, nickel-based alloys and stainless steels under the saturated pool boiling of water. As shown in Table 5, the pool boiling CHF values of as-received/post-CHF oxidized FeCrAl-C26M are greater than other 5 nuclear cladding materials. However, FeCrAl- B126Y and FeCrAl-B136Y have the pool boiling CHF values lower than that of Zircaloy 4. It should be noted that the as-received samples are less resistant to the boiling crisis than the post-CHF surface of tested samples that are oxidized by the CHF occurrence. This agrees with the pool boiling CHF experimental results of oxidized sample mentioned in Ref. [33]. The mechanistic rationales behind the pool boiling CHF enhancement by oxide layer are attributed to several dominant factors. For example, the metallic oxides give the surface wettability promotion, and the denser nucleate site density results from surface morphological features of oxide layers, and the oxide layer at the boiling heat transfer interface can induce the thermal-physical properties variations, as reported in the literature. As for which mechanism takes a dominant role, insofar, there is no consensus view, because three aforementioned mechanisms act together up on the CHF re-occurrence on the post-CHF surfaces. Table 6 lists out the pool boiling experimental CHF values of three Inconel and two titanium alloys. Inconel 600

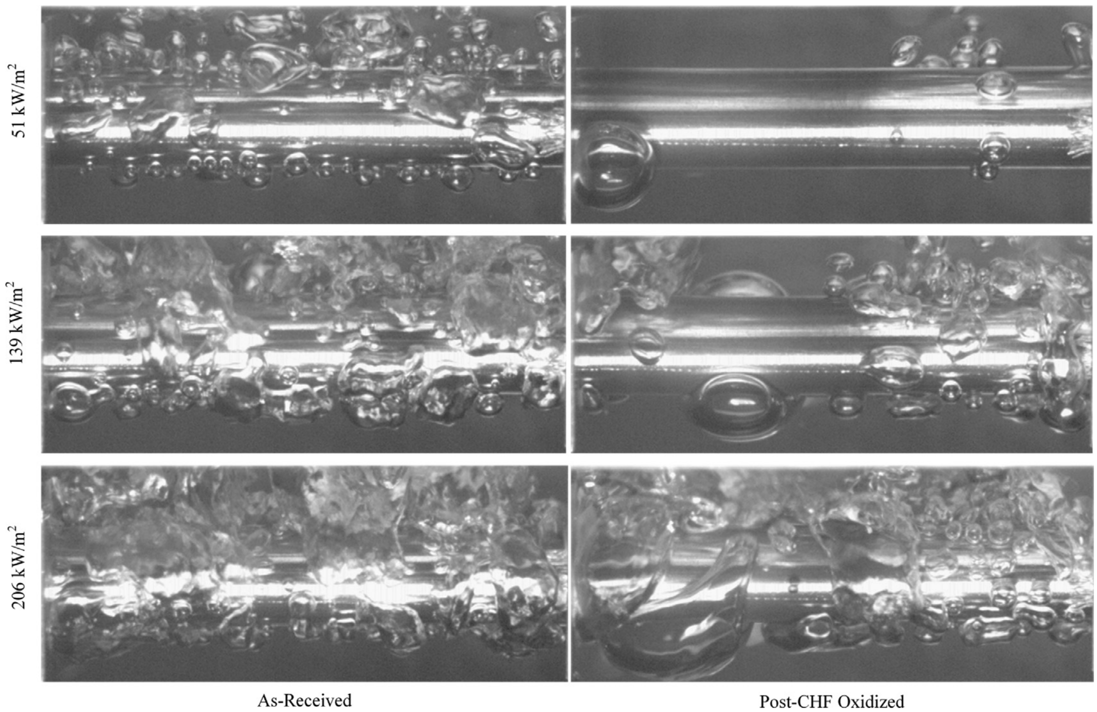


Fig. 14. Bubble behaviors of as-received and post-CHF FeCrAl-B126Y .

Table 5
Experimental pool boiling CHF of three Zircaloys and three FeCrAl alloys.

Material	OD/WT(mm)	Surface morphology	CHF (kW/m^2)	ΔCHF (kW/m^2)
Zircaloy 4	9.525/0.508	As-Received	567.3	±17.25
		Sandblasted by Grit 180	607.89	±18.48
Zirlo	9.475/0.486	As-Received	419.32	±14.34
		Post-CHF Oxidized	549.48	±18.79
Zr 705	9.500/0.650	As-Received	417.41	±16.78
		Post-CHF Oxidized	521.41	±20.96
FeCrAl-C26M	9.474/0.353	As-Received	648.25	±13.16
		Post-CHF Oxidized	858.2	±17.42
FeCrAl-B126Y	9.487/0.386	As-Received	420.55	±8.54
		Post-CHF Oxidized	499.21	±10.13
FeCrAl-B136Y	9.499/0.374	As-Received	380.96	±7.73
		Post-CHF Oxidized	549.5	±11.15

Table 6
Experimental pool boiling CHF of Inconels and titaniums.

Material	OD/WT (mm)	Surface morphology	CHF (kW/m^2)	ΔCHF (kW/m^2)
Titanium-2	9.525/0.889	Sandblasted by Grit 180	748.52	±37.58
Titanium-9	9.525/0.508	As-Received	398.18	±19.99
		Post-CHF Oxidized	611.71	±30.71
Inconel 400	9.525/0.508	Sandblasted by Grit 180	763.86	±15.43
	9.525/0.889	Sandblasted by Grit 180	645.24	±16.91
Inconel 625	9.525/0.508	As-Received	588.62	±11.89
Inconel 600	9.525/0.508	As-Received	646.04	±13.05
	9.525/0.7112	As-Received	452.23	± 9.14
	9.525/0.889	As-Received	419.32	±10.99
		Post-CHF Oxidized	571.49	±14.97

has a comparable pool boiling CHF with FeCrAl-C26M despite of their material differences. However, Inconel 400 and two titanium alloys have higher pool boiling CHF values than Inconel 600 and 625 because of their rough surfaces sandblasted by the sandpaper (grit 180)[2]

In Table 6, it is shown that the increasing of Inconel 600 wall thickness can result in the decreasing of pool boiling CHF. This is also observed in the pool boiling CHF experimental results of SS 316 with 5 different wall thickness (See Table 7). However, for experimental results of SS 304, the parametric trend of pool boiling CHF with respect to increasing of the wall thickness firstly decreases, then increases and finally decreases. The local maximum of SS 304 pool boiling CHF achieves at the wall thickness of 0.02 inch. The pool boiling CHF results of horizontally placed tubes contradict with that the pool boiling CHF of

[2] Titanium alloys and Inconel 400, their outside surfaces have a thin layer of oxide. To improve the heating efficiency of electrical power, this oxide layer is removed by using sandpapers.

Table 7
Experimental pool boiling CHF of stainless steels with various wall thicknesses.

Material	OD/WT (mm)	Surface morphology	CHF (kW/m^2)	ΔCHF (kW/m^2)
SS 304	9.525/0.254	As-Received	564.96	±11.41
		Post-CHF Oxidized	723.66	±14.62
	9.525/0.381	As-Received	441.38	±8.92
		Post-CHF Oxidized	922.25	±18.63
	9.525/0.508	As-Received	559.27	±11.30
		Post-CHF Oxidized	620.43	±12.53
	9.525/0.7112	As-Received	525.82	±10.62
	9.525/0.889	As-Received	519.5	±10.50
SS 316	9.525/0.254	As-Received	551.94	±11.50
	9.525/0.381	As-Received	518.93	±10.48
	9.525/0.508	As-Received	485.51	±9.82
		Post-CHF Oxidized	673.27	±13.60
	9.525/0.7112	As-Received	460.54	±9.30
	9.525/0.889	As-Received	427.9	±8.64
SS 321	9.525/0.508	As-Received	556.65	±11.24
		Post-CHF Oxidized	751.47	±15.18
SS 347	9.525/0.508	As-Received	434.63	±8.78
		Post-CHF Oxidized	564.03	±11.39

horizontally-place plates increases over the increasing of plate wall thickness.

4.2. Effects of wall thickness

The heat transfer scheme difference between the cladding tube and the square plate is that the cladding tube can transfer the heat to the ambient water radially at the full view of 360° while for the plate heater, the heat is transferred only from the one side of wall to the surrounding water. This heat flow direction difference may account for why the effect of tubing wall thickness on pool boiling CHF is different from the effect of plate wall thickness.

Speaking from the heat conduction perspective, the effect of tubing wall thickness on pool boiling CHF might be explained from the role of thermal capacity in heat conduction. To further support the analyses of heat conduction, the thermally equivalent diameter (D_t) is proposed based on the isothermal distribution of high-temperature tolerant cement filler

$$D_t = 2\sqrt{\mathrm{ODWT} - \mathrm{WT}^2} \tag{2}$$

where D_t is interpreted as the diameter of a fictitious wire heater releasing the same amount of heat as the tested tubing samples, OD and WT are the outer diameter and wall thickness of tubing samples respectively. *Thermal Penetration Depth:* In analogy to the transient heat conduction between semi-infinite bodies, this study adopts the concept of thermal penetration (diffusion) depth to characterize the effects of wall thickness on the pool boiling CHF, defined as follows,

$$\delta(t_d) = 3.64\sqrt{\alpha_s t_d} \tag{3}$$

where t_d is the bubble departure time, and α_s is the thermal diffusivity of cladding material. This new physic indicator quantifies the depth to which bubble effects propagate within a solid medium. t_d can be calculated by the empirical correlations. Using the thermal diffusivity at the cladding surface temperature before CHF occurrence, the thermal diffusion length can be calculated, for SS 304, $\delta(t_w)_{SS304} = 0.059$ inch and for SS 316, $\delta(t_w)_{SS316} = 0.064$ inch. According to the mechanistic definition of thermal diffusion length Eq. (3), it is not a valid approach to treat the cladding materials as semi-infinite sloid if the wall thickness is less than the thermal diffusion length. As pointed by Bruder et al. [34], the wall thickness increasing leads to better thermal activity of cladding material and further results in higher CHF. This is validated with the pool boiling CHF experimental results of square plate heater. However, given that the wall thickness is comparable with the thermal diffusion length, the impacts of wall thickness on the transient heat transfer across the wall thickness can not be neglected within a complete bubble ebullition cycle. The boiling fourier number Fo^* is defined as follows

$$Fo^* = \frac{4\alpha_s t_d}{(D_t)^2} \tag{4}$$

to quantify the dimensionless transient heat conduction induced by bubble ebullition. and the transient surface heat flux across the wall thickness can be simply approximated by the transient heat transfer of constant heat flux, such that

$$q'' \propto g(D_t) = \frac{1}{D_t\left[2Fo^* + 0.25 - 2\sum_{n=1}^{\infty}\frac{\exp(-\xi_n^2 Fo^*)}{\xi_n^2}\right]} \tag{5}$$

where the ξ_n is the nth root of the first-order Bessel function ($J_1(\xi_n) = 0$).

Fig. 19 clearly demonstrates that under influential impacts of bubble boiling behaviors, the progressive increasing of wall thickness can deteriorate surface heat flux, also have negative influential impacts on pool boiling CHF. This competes with the thermal activity enhancement by the wall thickness to the pool boiling CHF. Those two competing mechanisms explain our pool boiling CHF results in Table 7.

4.3. Effects of thermal-physical properties

Although temporal-spatial variations of working fluid phase change are a pronounced mechanism of nucleate boiling heat transfer before dry spots, the interfacial temperature between the solid region and the vapor region could be physcially explained by the suddenly quenching phenomenon of two bodies, one of which undergoes the phase change [35]. Loulou and Delaunay [35] derived the analytical expression of the interfacial temperature between two semi-infinite bodies as follows,

$$\frac{T_1 - T_2}{T_s - T_2} = \frac{\sqrt{(k\rho c_p)_2}}{\sqrt{(k\rho c_p)_1}} + 1, \tag{6}$$

where T_s is the interfacial temperature, T_1 and T_2 are the initial temperatures of the body 1 and the body 2 respectively before contacting, $\sqrt{k\rho c_p}$ is called the thermal effusivity. The physical interpretation of thermal effusivity is referred to as thermal inertia representing the material ability of exchanging heat with its surroundings. A higher thermal effusivity can allow nucleation sites of a material to be thermally activated in a faster manner. At the occurrence moment of CHF, the heat flow is severely impeded by

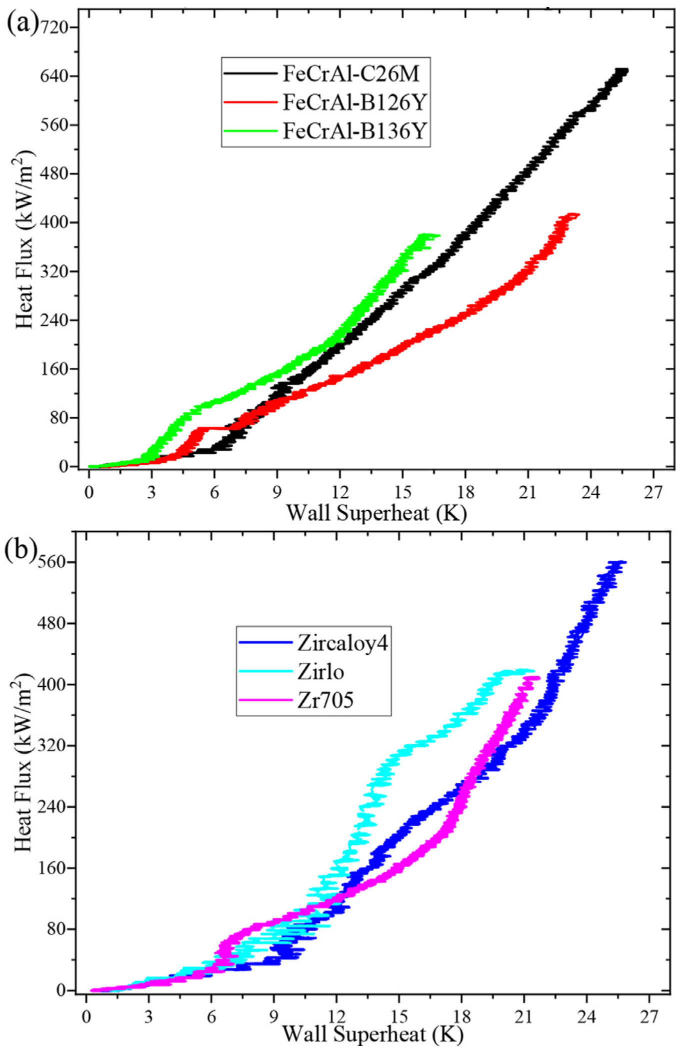


Fig. 15. Pool boiling curves of three FeCrAl alloys and three Zircaloys in the saturated water.

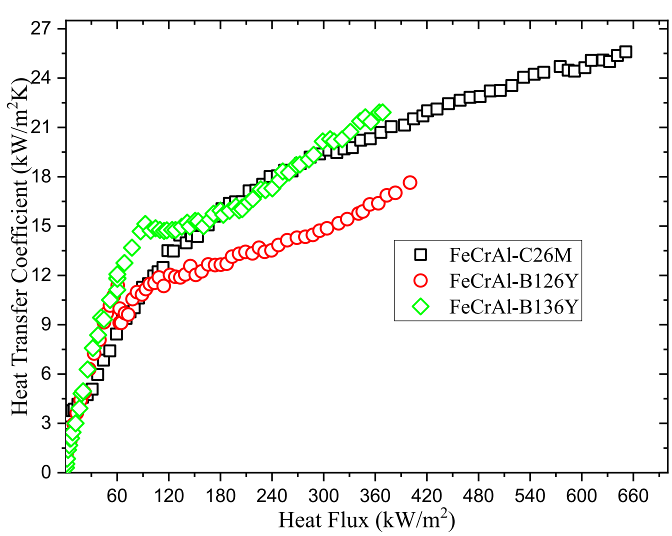


Fig. 16. Pool boiling heat transfer coefficients of three FeCrAl alloys, C26M, B126Y and B136Y.

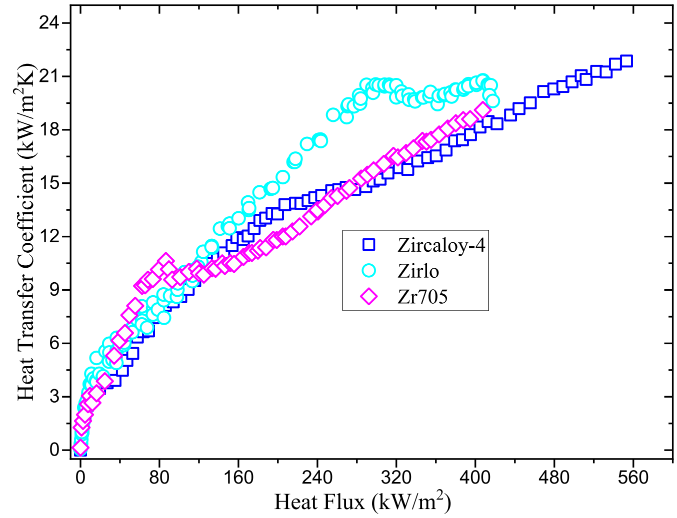


Fig. 17. Pool boiling heat transfer coefficients of three Zircaloys, Zircaloy 4, Zirlo and Zr 705.

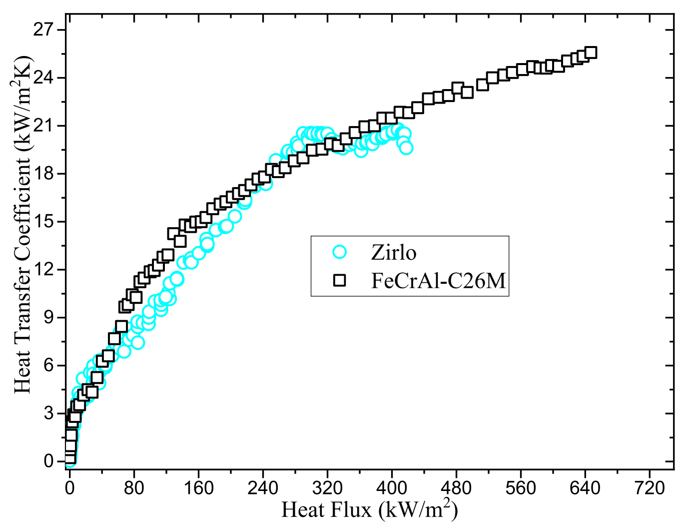


Fig. 18. Pool boiling heat transfer coefficients comparison between Zirlo and FeCrAl-C26M.

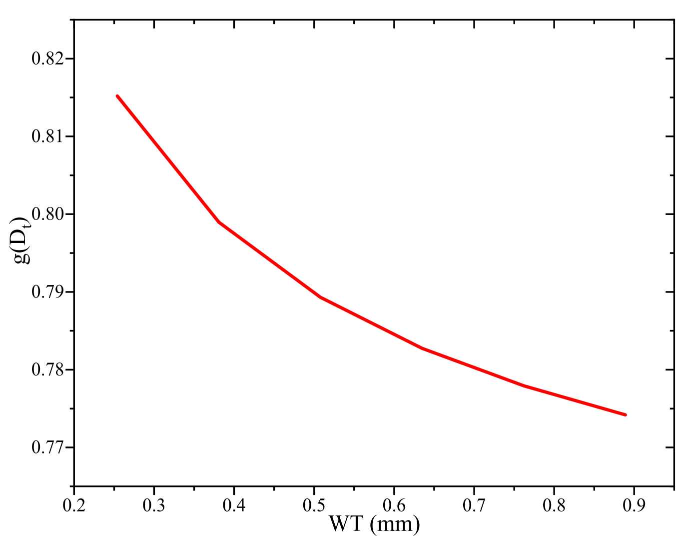


Fig. 19. Impacts of cladding wall thickness on the dependence of surface heat flux on Fo^{*}.

the vapor blanket of boundary layer separation model, the bubbly layer of bubble crowding model, the dry patch of sublayer dryout model, or the dry wall of interfacial lift-off model [36]. Besides, the average thickness of these vapor obstacles is at the small scale and the temperature drop across the thickness of tested tube could be negligible.

Basu et al. [37] indicated that the nucleation site density is proportional to the exponent of wall superheat. Under the saturated flow boiling, the wall superheat is the temperature drop between the wall and the working fluid. From this standpoint, the total heat flux across the wall could be partially dependent on the difference of thermal effusivity between the working fluid and the solid wall. Therefore, CHF could be correlated with thermal effusivities of working fluid and wall material as follows,

$$q''_{cr} \sim \left(\frac{\sqrt{(k\rho c_p)_{wall}}}{\sqrt{(k\rho c_p)_{fluid}}} + 1 \right). \tag{7}$$

Eq. (7) implies thermal effusivity ratio has impacts on pool boiling CHF instead of pool boiling CHF proportional to thermal effusivity.

The primary rationale behind the surface temperature overshoot is the local variation of the surface temperature. This transient temperature change can be characterized by the thermal diffusivities of wall material and working fluid because the importance of thermal diffusivity quantifies the heat transfer rate of a material from hot spots to cold spots. A wall material of higher thermal diffusivity can promote greater time-wise variation of surface temperature around nucleation sites, resulting in an earlier occurrence of CHF. On the other hand, the working fluid with higher thermal diffusivity such as nanofluids can lead to better heat dissipation from hot spots. Therefore, CHF is dependent on the relative heat dissipation between working fluid and heat surface material as follows,

$$q''_{cr} \sim \frac{\left(\frac{k}{\rho c_p}\right)_{fluid}}{\left(\frac{k}{\rho c_p}\right)_{wall}}. \tag{8}$$

Eq. (8) implies thermal diffusivity ratio has impacts on pool boiling CHF instead of pool boiling CHF proportional to thermal diffusivity. The mechanistic rationale behind Eq. (7) is that the heat quenching is dominated by thermal effusivity while thermal diffusivity has a direct dominant effect on bubble departure frequency in the heat flux partitioning model [38].

Another evident interaction at the interface between the solid and the liquid layer is the hydraulic attachment of working fluid to the solid because of the tradeoff balance between adhesive force and cohesive force, which is the surface wettability as a result from the intermolecular interactions. The surface wettability was experimentally proven the influential impacts on CHF of pool boiling [39] and flow boiling [40]. The effects of surface material wettability on CHF are often quantitatively and qualitatively analyzed by the measure of the dynamic receding contact angle θ_r [31],

$$q''_{cr} \propto (1+\cos\theta_r)\left(\frac{2}{\pi} + \frac{\pi}{4}(1+\cos\theta_r)\right)^{0.5}. \tag{9}$$

Hereby pool boiling CHF is mechanistically proportional to the surface wettability factor of Kandlikar model. But other alternatives to Eq. (9) may also help correlate the impacts of surface wettability on pool boiling CHF, such as the Liao's model [41]. It should be noted that the contact angle is also determined by surface morphology including roughness and microstructure. To minimize the effect of the surface morphological characteristics on CHF, the heat transfer surface of tested material should be smooth and plain.

The discussions above support that CHF is affected by thermal-physical properties and hydraulic behaviors of materials. This study aggregates the discussed parameters above together to define the nucleate boiling surface stability (NBSS) of heat surface material, defined as follows,

$$\mathrm{NBSS} = \left(\frac{\sqrt{(k\rho c_p)_{wall}}}{\sqrt{(k\rho c_p)_{fluid}}} + 1 \right) \frac{\left(\frac{k}{\rho c_p}\right)_{fluid}}{\left(\frac{k}{\rho c_p}\right)_{wall}} (1+\cos\theta_r)\left(\frac{2}{\pi} + \frac{\pi}{4}(1+\cos\theta_r)\right)^{0.5}. \tag{10}$$

The physical interpretation of nucleate boiling surface stability is to quantify the ability that the material of heat transfer surface can sustain the nucleate boiling of working fluid.The better nucleate boiling surface stability can much delay the pool boiling CHF occurrence and lead to higher CHF. It should be noted that the outer surface temperature is calculated to evaluate the solid thermal-physical properties based on the steady-state 1-D heat conduction model while the liquid thermal-physical properties are evaluated by the saturation temperature.

Gaertner [42] had shown that a large fraction of wetted surface was still observed right before pool boiling CHF occurrence, and a vapor blanket started from the dry patches coalescence to the growth expansion covering the entire surface. This could not be actually reflected by Eq. (10). However, Eq. (10) aggregates thermal-physical properties together with surface wettability to better indicate how heaters affect pool boiling CHF. Although thermal-physical material properties of tested tubes vary with the increasing of temperature, many metallic and nonmetallic materials show the monotonic increasing parametric trends of those thermal-physical properties with respect to temperature range of nucleate boiling regime. Characterization at the saturation temperature of working fluid could still provide some insights on impacts of thermal-physical material properties on CHF. To conveniently elaborate the impacts of thermal-physical material properties on CHF, the right hand sides of Eqs. (7)–(9) are respectively defined as the thermal effusivity ratio, the thermal diffusivity ratio and the wettability. It should be noted that the mechanistic discussions are elaborated based on claddings instead of heating elements.

Effects of thermal-physical material properties on pool boiling CHF are more pronounced without the impacts of forced convection [13,43]. This study collects pool boiling CHF experiments of water and FC-72 on 10 mm × 10 mm heat plates made of different tested materials from published literatures. The CHF experimental data points from Ref. [13] are used to evaluate thermal-physical properties in water saturated pool boiling CHF. The CHF experimental data points of FC-72 pool boiling are respectively collected from Al-6061/$AlSi_{10}Mg$ [43], Silicon [44], copper [45], Steel 1010 [46], Silica [47], Aluminum [48], and SS 316 [49].

The general trend of CHF shows an increasing parametric behavior with respect to thermal diffusivity ratio, as can be observed in Figs. 20(a) and 21(a). This further verifies that a higher thermal diffusivity of material results in a lower CHF. But the thermal diffusivity cannot fully explain the effects of heater materials on CHF alone because CHF can still vary significantly on the materials with similar thermal diffusivities (Fig. 20(a)) and vice versa (Fig. 21(a)).

The parametric behaviors of CHF with respect to the thermal effusivity ratio demonstrate the similar patterns in pool boiling of water (Fig. 20(b)) and FC-72 (Fig. 21(b)). This suggests that the way of heater materials affecting the occurrence of CHF should behave similarly in different working fluids. Despite of the similar effects of heater materials on pool boiling CHF of different working fluids, the parametric trend of CHF with respect to thermal effusivity ratio shows highly nonlinear behaviors. As for the effect of material wettability on CHF, the CHF model proposed by Kandlikar suggested CHF could be enhanced by the heater materials with higher wettability. However the experimental results in Figs. 20(c) and 21(c) contradicted the predicted parametric trend of CHF against mate-

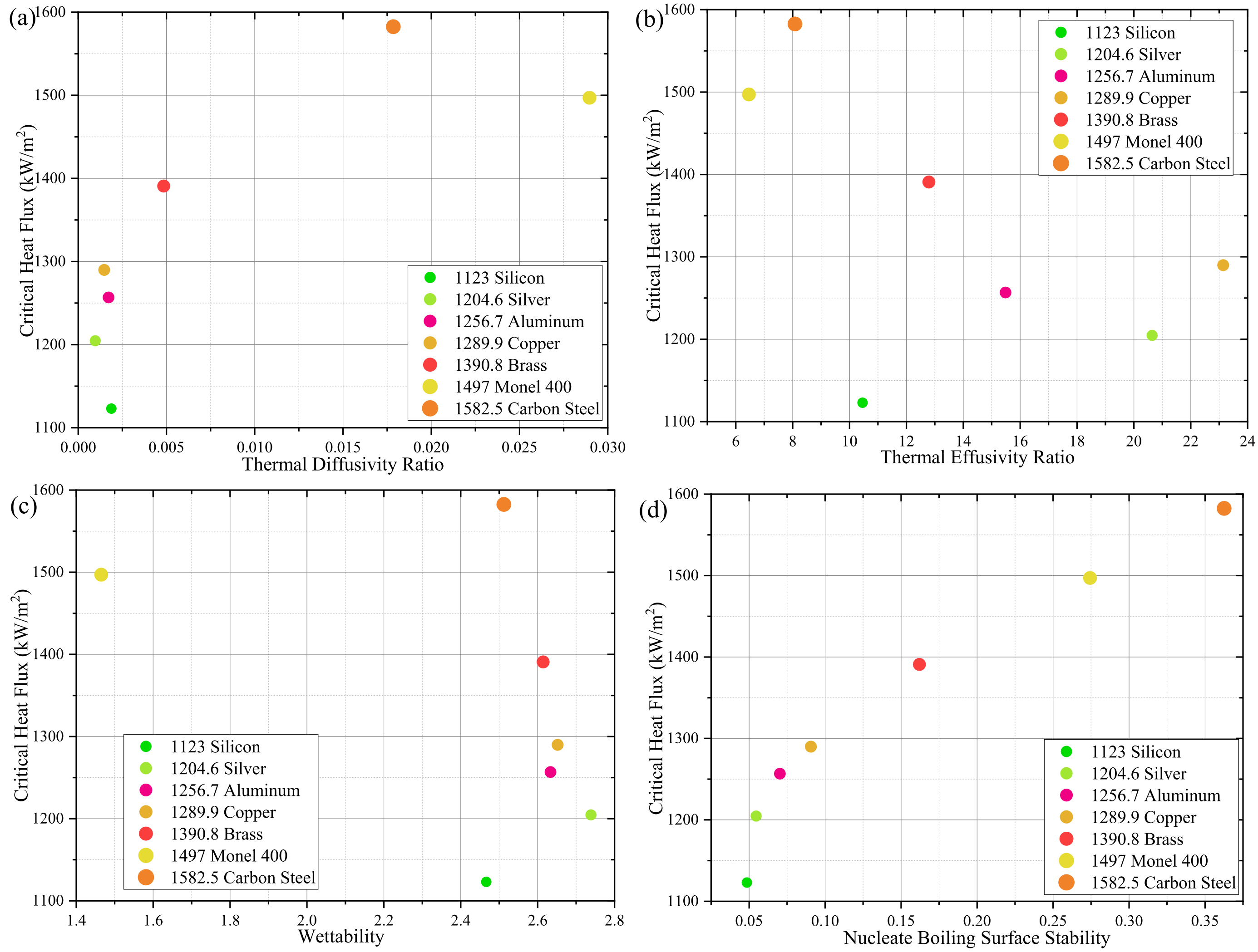


Fig. 20. Water pool boiling CHF as a function of thermal-physical properties of common materials,(a) thermal diffusivity, (b) thermal effusivity, (c) wettability, and (d) nucleate boiling surface stability.

rial wettability in the Kandlikar's model. It is because the effects of material wettability on CHF was correlated and incorporated in the Kandlikar's model while the influences of thermal diffusivity and thermal effusivity were not evaluated to the triggering mechanisms of CHF.

As the aforementioned discussions, thermal diffusivity, thermal effusivity and material wettability are incapable of addressing the effects of heater materials on CHF alone. The new proposed thermal-physical material parameter, "nucleate boiling surface stability", aggregates their contributions to the triggering mechanisms of CHF. As can be observed in Figs. 20(d) and 21(d), the nucleate boiling surface stability of heater material can render a satisfactory rationale to the obvious CHF difference between different tested materials with an apparent increasing relation. It should be noted that both pool boiling experimental CHF results in Figs. 20 and 21 are obtained upon the horizontally-placed square claddings.

However, the proposed nucleate boiling surface stability is incapable of parameterizing the pool boiling CHF results of cladding tube materials (See Fig. 22) but still provides a better parametrization than other three indicators. This could be possibly rationalized by the heating-element tube boiling due to a different heat conduction mechanism. Boiling heat transfer of claddings would be different from that of heating elements because of heat flux generation mechansims. Claddings are used to encase volumetric heat sources and protect against corrisions and thermal energy is transferred to the boiling interface through heat conduction across the cladding layer such as the fuel-cladding entity of LWRs. However, heating elements generate thermal energy within their solid media and release heat flux at their outer surfaces. This direct-heating method coupled with boiling cylindrical surface could be different from the indirect thermal energy across the cladding.

It should be noted that the experimental results in Fig. 20 are obtained from the horizontally placed upward-facing square plates. Their surface orientation angles are 0°. According to the Liao's CHF model of saturated pool boiling of water with incorporation of surface orientation [41], the predicted CHF is 1075.68 kW/m^2. On the one hand, pool boiling CHF difference between Liao's modeling and experimental results could be attributed to the thermal-physical properties. On the other hand, the extreme CHF case of water pool boiling is the horizontally placed downward-facing plates, and its predicted CHF is 249.75 kW/m^2. However, for pool boiling on horizontally-placed tubes, surface contact angles vary from 0 to 180°. In comparison with the pool boiling on the horizontally placed downward-facing plates, bubbles are much less likely to accumulate on the bottom of horizontally-placed tubes. Our experimental observations showed that pool boiling CHF first occurs on

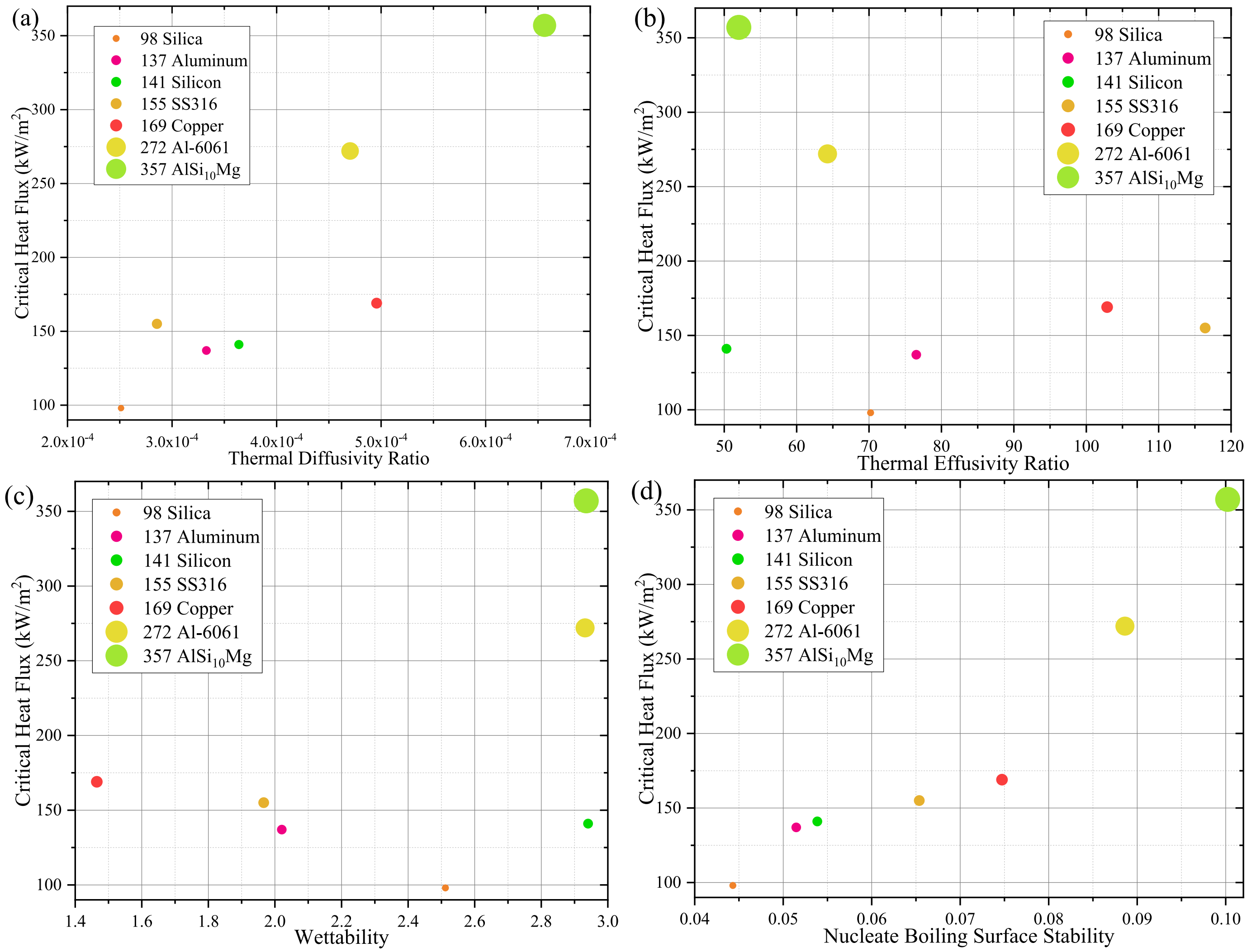


Fig. 21. FC-72 pool boiling CHF as a function of thermal-physical properties of common materials,(a) thermal diffusivity, (b) thermal effusivity, (c) wettability, and (d) nucleate boiling surface stability.

the bottom surface, and then expands to the top surface of horizontally placed tubes. This speaks to that pool boiling CHF values of horizontally-placed tubes are supposed to be higher than that of horizontally placed downward-facing plates. Given that there are enhancement contributions of thermal-physical properties that are less dominant than surface orientation, it is mechanistically acceptable that pool boiling CHF experimental results of horizontally-placed upward facing plates in Fig. 20 are two to three folds higher than our experimental results of horizontally-placed tubes in Fig. 22. Besides the impact of surface inclination angles on pool boiling CHF, there are also other two primary potential factors affecting pool boiling CHF, i.e., system pressure and Taylor instability. The system pressure of our pool boiling chamber is 84 kPa while most of pool boiling CHF experimental results were obtained under the standard atmospheric pressure of 101 kPa. This pressure difference can partially explain the relatively low CHF values on horizontal tubes. The other major contribution to low CHF values from our experimental pool boiling experiments could possibly be the Taylor instability of bubble dynamics on horizontally-placed tubes. Bubbles from the upper half and lower half of tube tend to merge and form vapor columns [18]. In comparison with horizontally-placed upward-facing plates, the number of vapor columns is less than that on the horizontally-placed tubes for the same surface area. This also accounts for lower surface heat flux triggering vapor columns coalescence, and further results in the lower CHF results on horizontally-placed tubes. Moreover, in this study, the experimental results of horizontally-place tube pool boiling CHF are scattered. Similar results were also reported by Rao and Andrews [18], as can be seen in Fig. 6 in Ref. [18].

4.4. Modeling assessments

Various mechanistic models or empirical correlations have been obtained for pool boiling CHF prediction of flat plates. However, limited experimental and modeling studies characterizing pool boiling CHF on horizontal cylinders [16,18,19]. These experimental and modeling results speak to that various hydrodynamics instabilities dominate pool boiling CHF triggering mechanisms using small diameter wires/rods and tubes. In this experimental study, the capillary length of saturated water is 2.521 mm. According to the criteria of three models [16,18,19], the outer diameters of the tested samples in Table 1 fall into the application range of Sun and Lienhard's model [16]. Sun and Lienhard [16] assumed that the Helmholtz-instable collapse of vapor columns was the primary driving mechanism behind pool boiling CHF occurrence on the horizontal cylinders because of the Rayleigh's capillary instability. They derived an analytical model to correlate the impacts of radius on pool boiling CHF of horizontally-placed rodlet heaters as

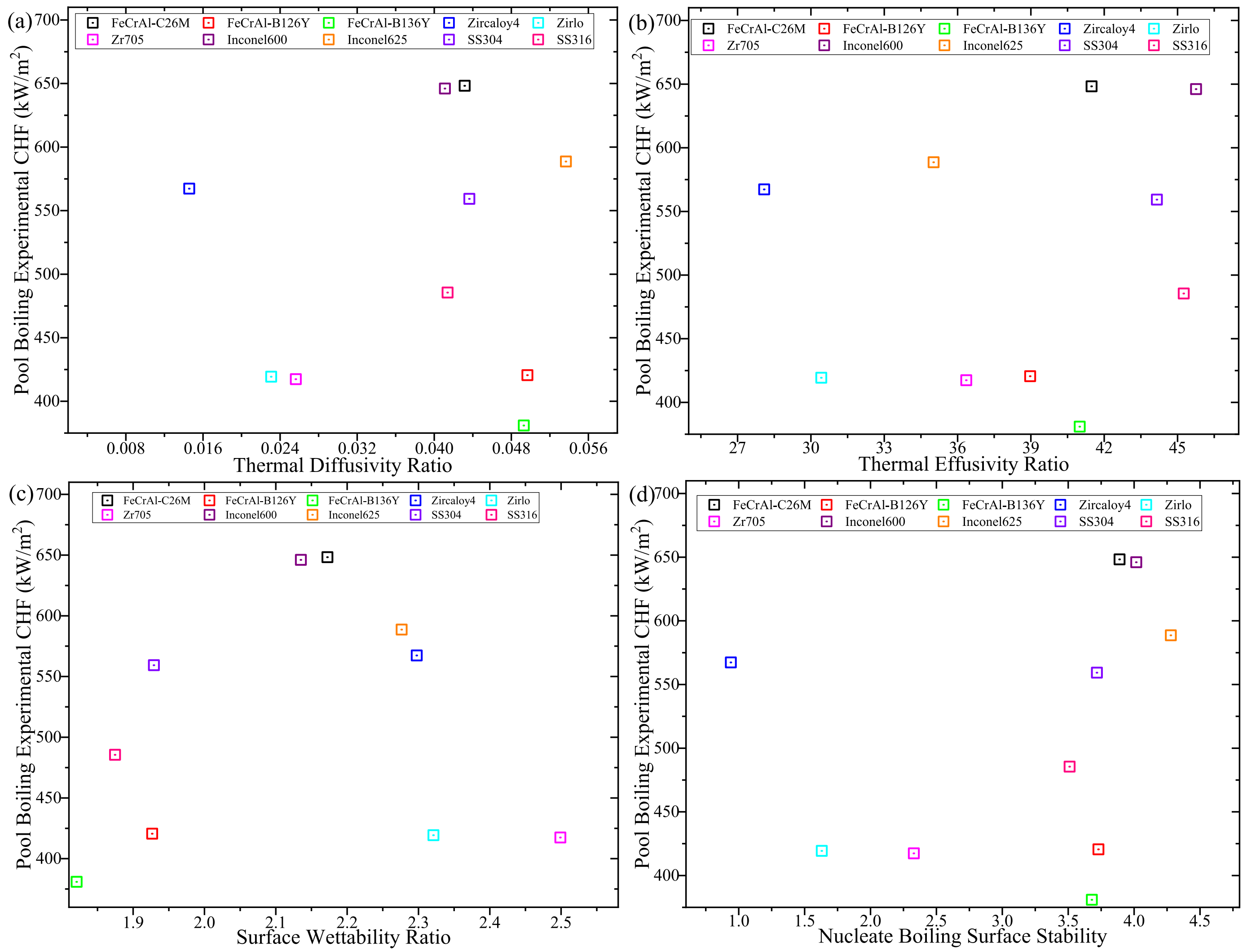


Fig. 22. Pool boiling CHF experimental results of horizontally-placed tubes assessed by four different physical indicators, (a) thermal diffusivity, (b) thermal effusivity, (c) wettability, and (d) nucleate boiling surface stability.

follows,

$$\frac{q''_{cr}(R)}{q''_{cr,\mathrm{F}}} = \begin{cases} \dfrac{6}{\pi^2\sqrt{3}}\dfrac{(R'+\omega)^{1.5}}{R'} & R'+\omega \le 4.28 \\ 1.233\dfrac{3^{\frac{3}{4}}}{\pi} & R'+\omega > 4.28 \end{cases} \tag{11}$$

where R' is the dimensionless radius defined the ratio of the heater radius R over the capillary length of working fluid L_c, ω is defined as a dimensionless vapor blanket thickness and $q''_{cr,\mathrm{F}}$ is the pool boiling CHF of infinitely large horizontal upward-facing flat plate. In Eq. (11), ω is calculated via an empirical equation as follows,

$$\omega = \begin{cases} [2.54R' + 6.48R'\exp(-3.44\sqrt{R'})]^{\frac{2}{3}} - R' & R' \le 3.47 \\ 0.233R' & R' > 3.47 \end{cases} \tag{12}$$

Noting that Eq. (12) is applicable to Eq. (11) if and only if $R' > 0.15$. Otherwise, pool boiling CHF is governed by another gravity-capillary instability. More details can be found in Ref. [16]. The dimensionless radii of the test samples are greater than 3.756 according to Eqs. (11) and (12). This implies that the pool boiling experimental CHF results in this study can be evaluated by the CHF models of horizontal upward-facing plates and corrected by a factor of $1.233\frac{3^{\frac{3}{4}}}{\pi}$. However in Eq. (11), $q''_{cr,\mathrm{F}}$ is evaluated by the Zuber pool boiling CHF model in Sun and Lienhard's study. To assess the effects of surface wettability and roughness on pool boiling CHF, $q''_{cr,\mathrm{F}}$ is evaluated by the Kim et al. [50]'s CHF model in this study as follows,

$$q''_{\mathrm{CHF}} = S\Delta H_{fg}\rho_g^{0.5}[\sigma G(\rho_l - \rho_g)]^{0.25}\,\frac{1+\cos\theta_r}{16}\left[\frac{2}{\pi} + \frac{\pi}{4}(1+\cos\theta_r) + \frac{4C\cos\theta_r}{1+\cos\theta_r}\left(\frac{\mathrm{Ra}}{\mathrm{SM}}\right)\right]^{0.5} \tag{13}$$

where C is the proportionality factor for the effects of roughness-enhanced capillary force on bubble departure dynamics (it is found to be 87.8 in Kim et al. [50]'s experimental data sets), and S is the surface accommodation factor (it is 0.811 in their study). In this study, Kim et al. [50]'s CHF model is used to predict the pool boiling CHF of horizontal upward-facing plates while Sun and Lienhard [16]'s model is adopted to correct predictive results of Kim et al. [50] for horizontally-placed tubes, as can be seen in Fig. 23.

As shown in Fig. 23, the predictive results of Sun and Lienhard [16]'s model are comparable with the experimental results of this study in terms of pool boiling CHF ranges. This potentially speaks to that Sun and Lienhard [16]'s pool boiling CHF model could resolve boiling pattern differences between plates and cylinders under the horizontally-placed configuration. However, both Sun and Lienhard [16]'s and Kim et al. [50]'s models don't account for pronounced impacts of thermal-physical properties and wall thick-

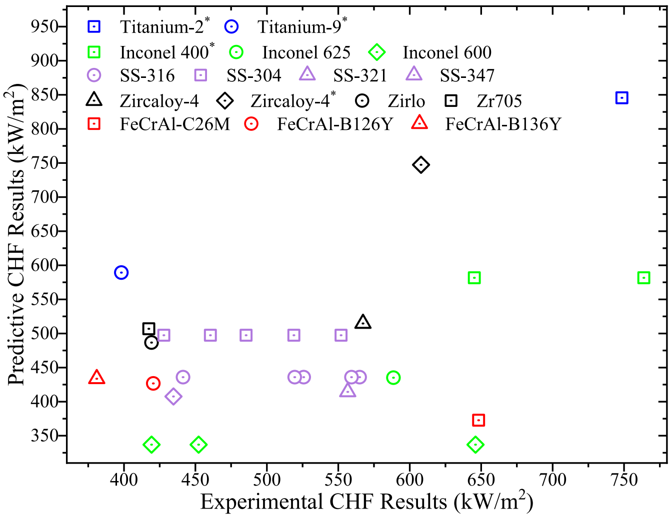


Fig. 23. Pool boiling experimental results assessment by using the Sun and Lienhard [16]'s and Kim et al. [50]'s CHF models noting that * means the sandblasted samples.

ness on pool boiling CHF. This mechanistically attribute to large CHF variations between predictive and modeling results. In light of this, the separate and integral experimental investigations are much needed to characterize the potential impacts of various factors on pool boiling and demystify the mechanistic rationales behind CHF occurrences.

5. Conclusions

In the pool boiling experimental investigations of FeCrAl alloys, Zircaloys, Inconel alloys and stainless steels, the experimental results demonstrate that FeCrAl-C26M has higher pool boiling CHF and NB-HTC than other cladding samples. This speaks to the thermal superpriority of FeCrAl-C26M in the LWR core loading. However, the effect of tubing wall thickness on pool boiling CHF covered in this study contradicts with the effect of plate wall thickness on CHF. This could result from the geometrical scheme of heat transfer mechanism of the tubing heater. On the other hand, the parametric trend of pool boiling CHF with respect to the tubing wall thickness agrees with the experimental results of pool boiling CHF on the wire heaters with different wire diameters.

More importantly, the pool boiling experimental CHF results obtained from various cladding materials also confirm that material thermal-physical properties can contribute significantly to the pool boiling CHF and NB-HTC difference. But how the thermal-physical properties mechanistically affect the pool boiling heat transfer still remain vague even though several physical indicators of material thermal-physical properties reported in literature are incapable of parameterizing the pool boiling CHF behaviors. Besides pool boiling CHF and NB-HTC, the material thermal-physical properties can also play a critical role in the bubble dynamics behaviors. On the cladding surfaces with similar surface roughness scales, Inconel 600 and 625 generate larger bubbles and have a smaller number of bubble crowds than SS 304 and 316.

In this pool boiling CHF study of various cladding materials, it is confirmed that the post-CHF samples have higher pool boiling CHF values than their as-received counterparts. However, such pool boiling CHF enhancement by oxide layer cannot be mechanistically rationalized by the present boiling theories reported in literature. By comparing the oxide layer thicknesses of different cladding materials, it is found that FeCrAl-C26M has the thinnest one among all the tested materials. This implies that FeCrAl-C26M has the best resistance to the oxidization corrosion resulting from pool boiling CHF occurrence.

Declaration of Competing Interest

Authors declare that they have no conflict of interest.

CRediT authorship contribution statement

Mingfu He: Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft. **Amir F. Ali:** Methodology, Validation, Writing – review & editing. **Minghui Chen:** Conceptualization, Resources, Funding acquisition, Writing – review & editing.

Data Availability

Data will be made available on request.

Acknowledgments

This material is based upon work supported by the U.S. Department of Energy Office of Nuclear Energy's Integrated Research Program under award number DE-NE-0008531. Authors are very thankful to Dr. Kurt A. Terrani of ORNL, Dr. Raul Rebak of GE, Dr. Stuart Maloy of LANL for their generous donations of cladding materials.

Appendix A. Steady-state radial temperature distribution of test section

The steady-state heat conduction model of the central high-temperature tolerant filler can be simplified as a one-dimensional radial version,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(k_c r\frac{\partial T}{\partial r}\right) = 0,\ \ 0 \le r \le \frac{\mathrm{OD}}{2} - \mathrm{WT} \tag{A.1}$$

where T is the radially spatial distribution of temperature of test section, k_c is the cement filler thermal conductivity dependent of T, and r is the region of interest. An alternative form of Eq. (A.1) is written as follows

$$\int_{T_o}^{T_i} k_c(T)dT = C\int_0^{\frac{\mathrm{OD}}{2}-\mathrm{WT}}\frac{dr}{r} \tag{A.2}$$

where C is an unknown constant, T_o is the centerline temperature of test section, T_i is the interface temperature between tubing material and cement filler, and $k_c(T)$ is usually given by an empirical correlation. To make the right-hand side of Eq. (A.2) mathematically reasonable, the only feasible solution is to allow C to be zero. This also speaks to that the heat flux at the centerline of test section is zero. Therefore, Eq. (A.2) can lead to that the temperature of cement filler is uniformly distributed from the centerline to the interface and it implies that the cement filler is isothermal. For a complete cycle of power incremental step, the steady-state heat conduction model of the test specimens could be written as follows,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(k_t r\frac{\partial T}{\partial r}\right) + q''' = 0,\ \ \frac{\mathrm{OD}}{2} - \mathrm{WT} \le r \le \frac{\mathrm{OD}}{2} \tag{A.3}$$

where k_t is the tubing material thermal conductivity dependent of T, q''' is the volumetric heat source. And the boundary condition of Eq. (A.3) at the interface can be given as follows,

$$q''\left(\frac{\mathrm{OD}}{2} - \mathrm{WT}\right) = 0 \tag{A.4a}$$

$$T\left(\frac{\mathrm{OD}}{2}-\mathrm{WT}\right)=T_i \tag{A.4b}$$

noting that T_i is measured by K-type thermocouples at four different axial locations. Under the assist of the differential transformation method, the outer surface temperature of test specimen can be computed based on the interface temperature measurement.

Appendix B. Measurement uncertainties

The test section releases heat flux via the direct joule heating mechanism. To protect the DC programmable power supply, the voltage-based control mode is adopted to approximate the power incremental of test section. Once the steady state of boiling system is reached, such state will be kept at least for 40 s and then the voltage across the test section will be increased by a step of 0.01 V. Given that the electrical resistance varies between different cladding materials, the power incremental step (ΔP) resulting from the voltage-based control mode should be considered and calculated as follows,

$$\Delta P \cong \frac{2V}{R_e}\Delta V \tag{B.1}$$

where V is the voltage drop across the test section, ΔV is the voltage incremental step (in this study, it is 0.01 V) and R_e is the electrical resistance of effective heated test section. The experimental heat flux uncertainty $\Delta q''$ can be calculated as follows,

$$\begin{aligned}\frac{\Delta q''}{q''} &\cong \frac{\Delta q}{q}-\left(\frac{\Delta \mathrm{OD}}{\mathrm{OD}}+\frac{\Delta \mathrm{HL}}{\mathrm{HL}}\right)\\ &\cong 2\frac{\Delta U}{U}-\left(\frac{\Delta \mathrm{OD}}{\mathrm{OD}}+\frac{\Delta \mathrm{HL}}{\mathrm{HL}}\right)\end{aligned} \tag{B.2}$$

where OD is the outer diameter of test specimens, ΔOD is the manufacturing tolerance of outer diameter, HL is the effective heated length of test specimens, and ΔHL is the manufacturing tolerance of heated length, U and ΔU are the voltage drop and its uncertainty for the test specimens. Besides the uncertainty contributed by the voltage-based control mode, there also exists the measurement uncertainty of voltage drop across the test specimens during sampling the data points. In light of this, the following criterion is adopted to quantify the uncertainty of test section voltage drop

$$\frac{\Delta U}{U}=\max\left(\frac{\Delta V}{V},\frac{\Delta u}{u}\right) \tag{B.3}$$

where V is the test section voltage when the CHF occurs on the outer surface of test specimens, $\frac{\Delta u}{u}$ is the measurement uncertainty of voltage sensor. The relative uncertainty of heat flux is calculated as follows,

$$\frac{\Delta q''}{q''} \cong \sqrt{4\left(\frac{\Delta U}{U}^{2}\right)+\left(\frac{\Delta \mathrm{OD}}{\mathrm{OD}}\right)^2+\left(\frac{\Delta \mathrm{HL}}{\mathrm{HL}}\right)^2} \tag{B.4}$$

in this study. And also the ΔHL/HL and $\Delta U/U$ is 0.01 according to the vendors. The ΔOD/OD and $\Delta V/V$ are respectively provided in Table 1 based on various test specimens.

Appendix C. Optical and SEM micrographs

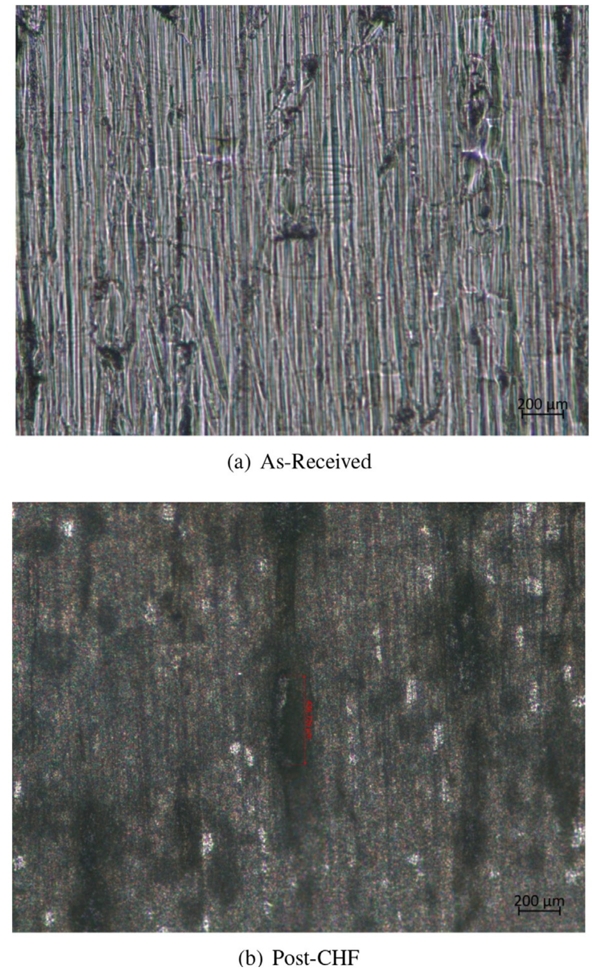
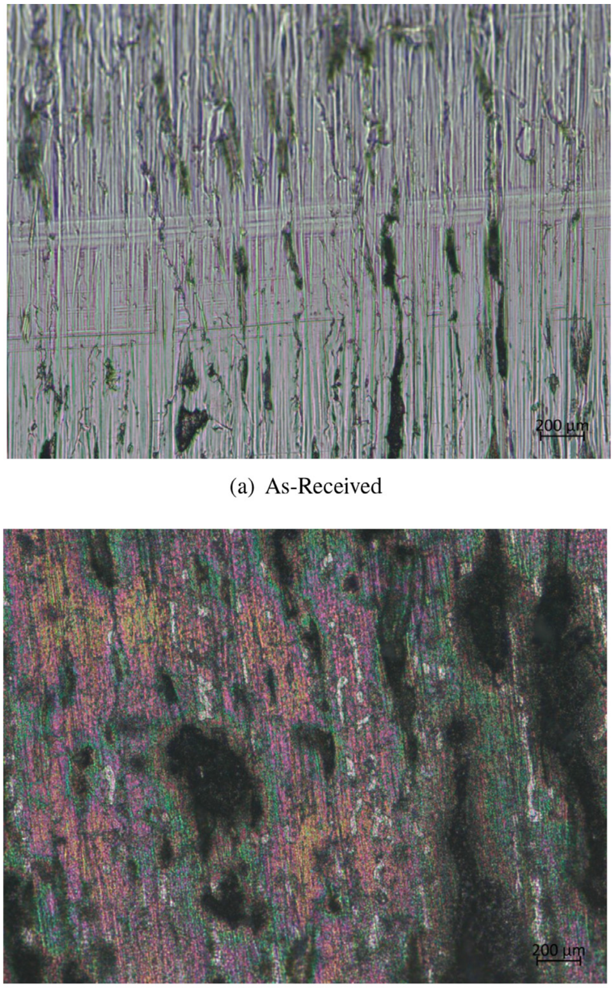


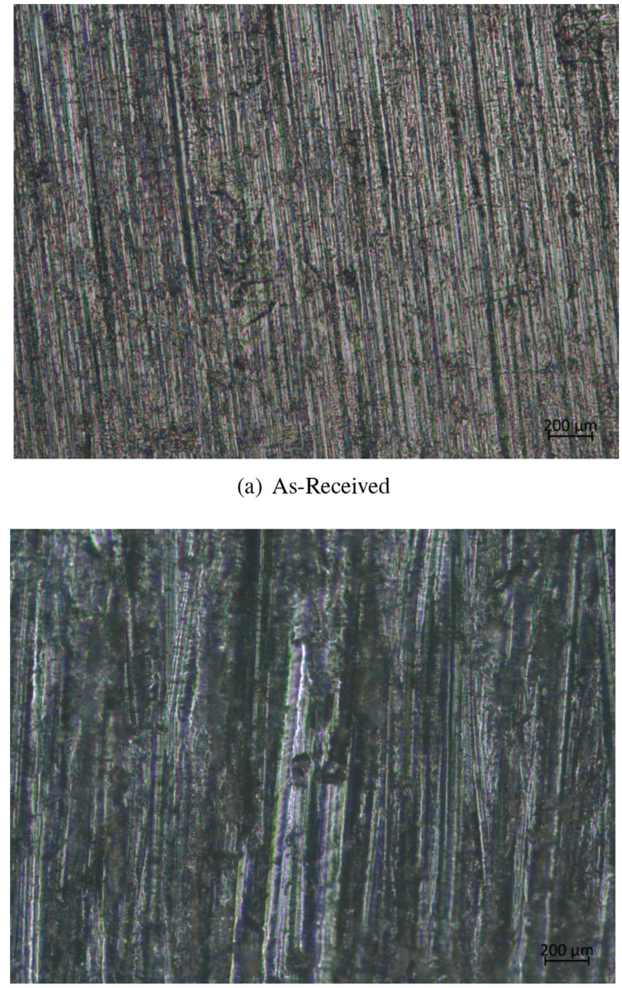
Fig. C1. Optical scan of FeCrAl-B126Y before and after boiling at the magnification of 20.



(a) As-Received

(b) Post-CHF

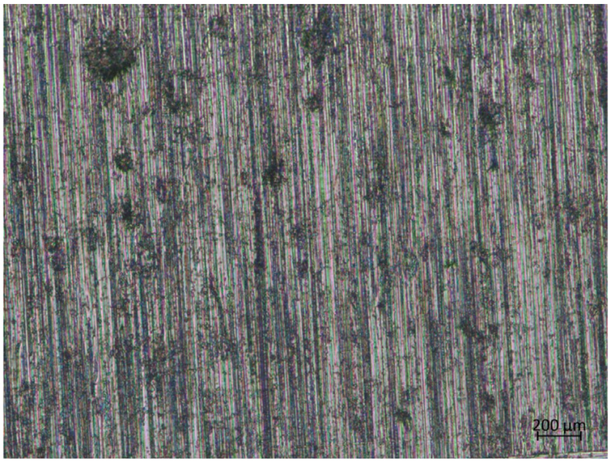
Fig. C2. Optical scan of FeCrAl-B136Y before and after boiling at the magnification of 20.



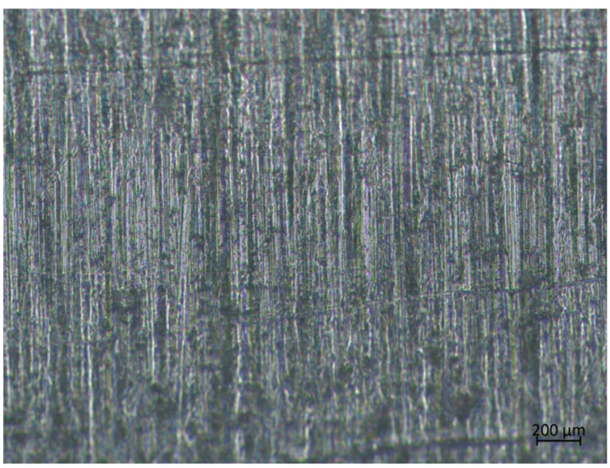
(a) As-Received

(b) Post-CHF

Fig. C3. Optical scan of Zircaloy 4 before and after boiling at the magnification of 20.

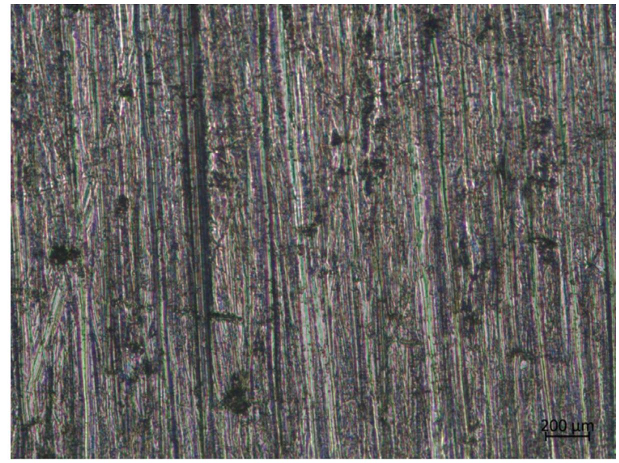


(a) As-Received

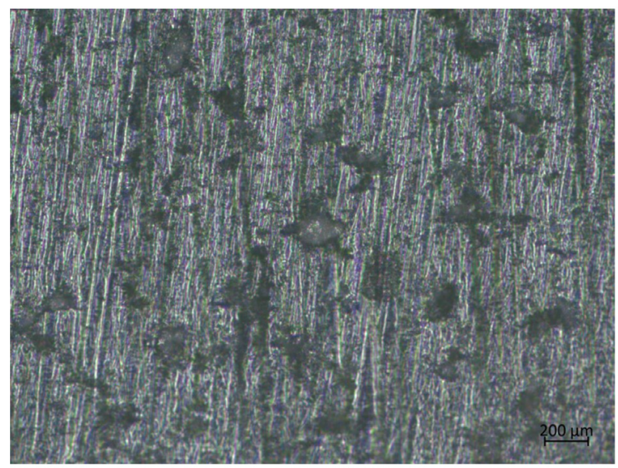


(b) Post-CHF

Fig. C4. Optical scan of Zirlo before and after boiling at the magnification of 20.

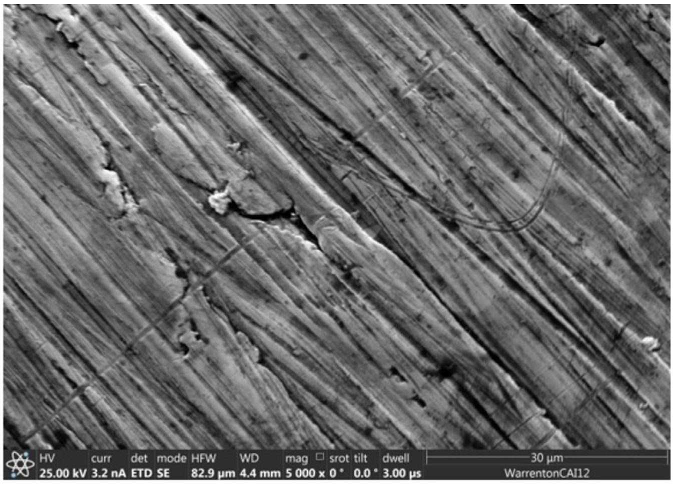


(a) As-Received



(b) Post-CHF

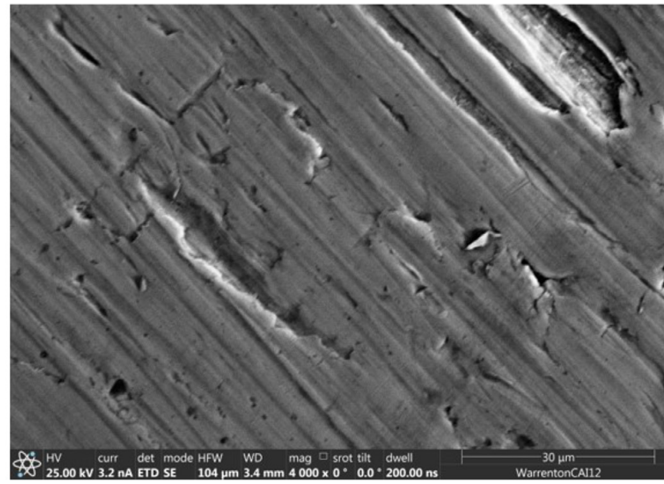
Fig. C5. Optical scan of Zr 705 before and after boiling at the magnification of 20.



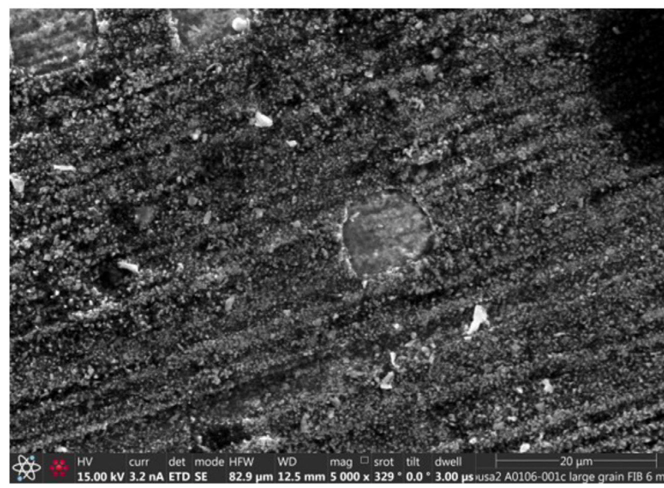
(a) As-Received

(b) Post-CHF

Fig. C6. SEM scan of FeCrAl-B126Y before and after pool boiling CHF experiment.

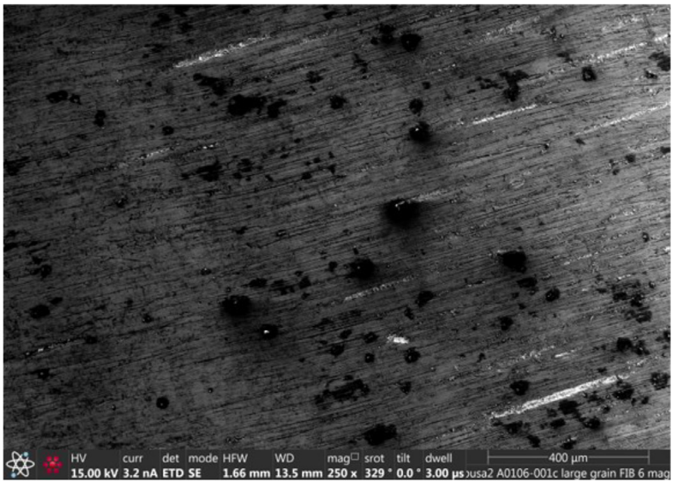


(a) As-Received

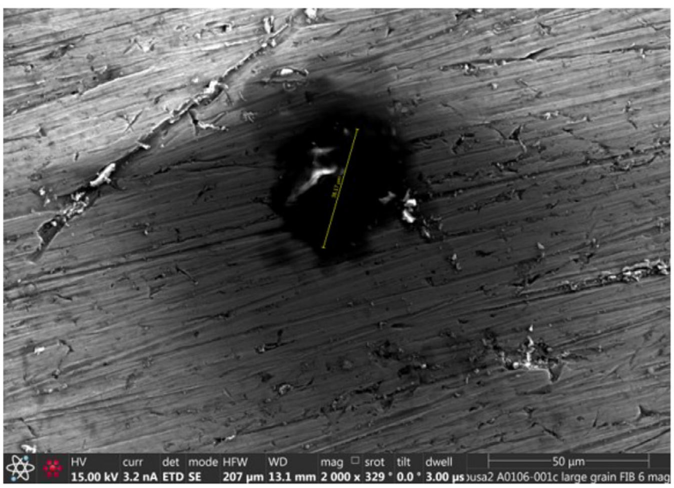


(b) Post-CHF

Fig. C7. SEM scan of FeCrAl-B136Y before and after pool boiling CHF experiment.

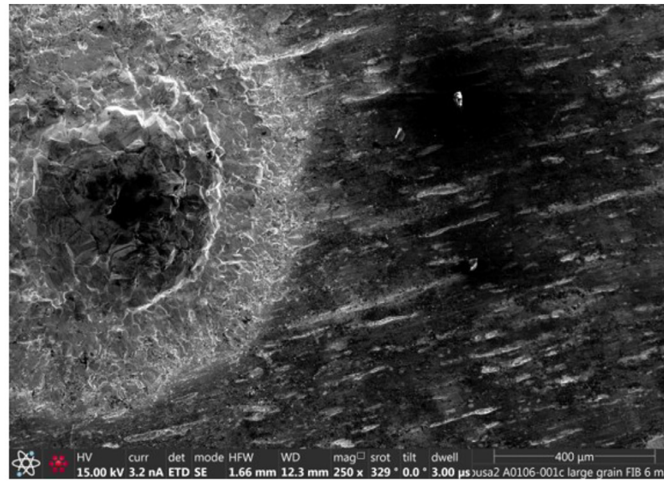


(a) At the magnification of 250

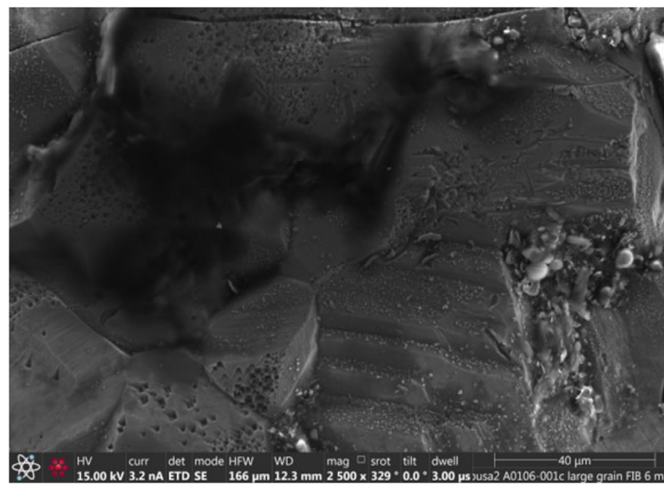


(b) At the magnification of 2000

Fig. C8. SEM scan for a typical feature of FeCrAl-C26M post-CHF surface.

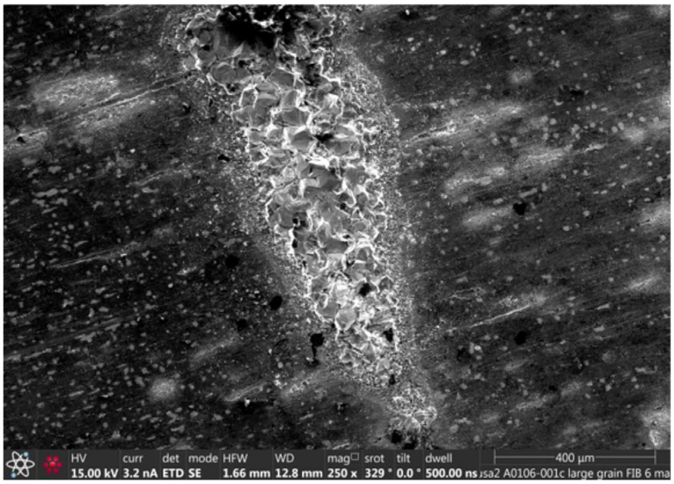


(a) At the magnification of 250

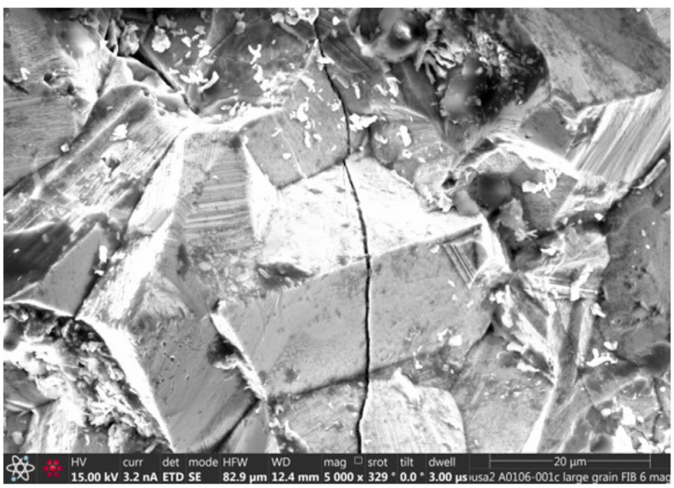


(b) At the magnification of 2000

Fig. C9. SEM scan for a typical feature of FeCrAl-B126Y post-CHF surface.

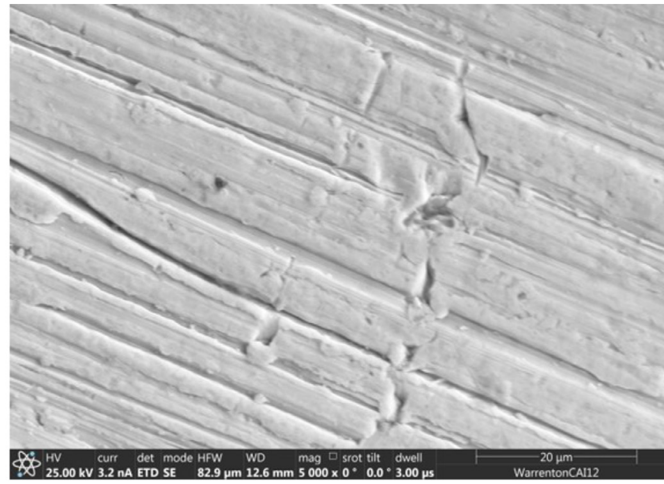


(a) At the magnification of 250

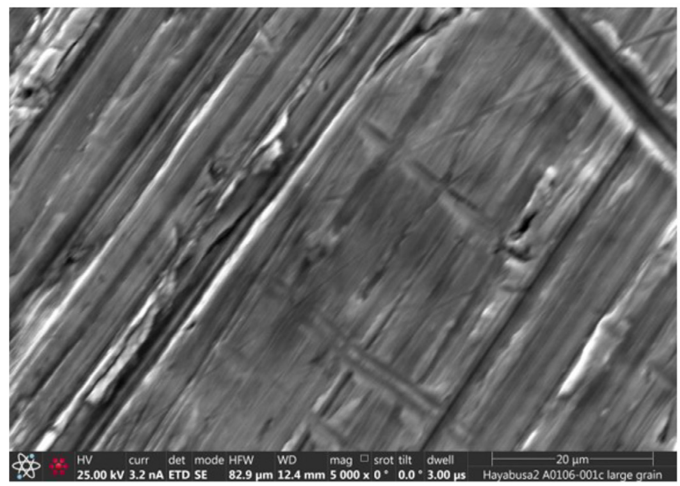


(b) At the magnification of 2000

Fig. C10. SEM scan for a typical feature of FeCrAl-B136Y post-CHF surface.

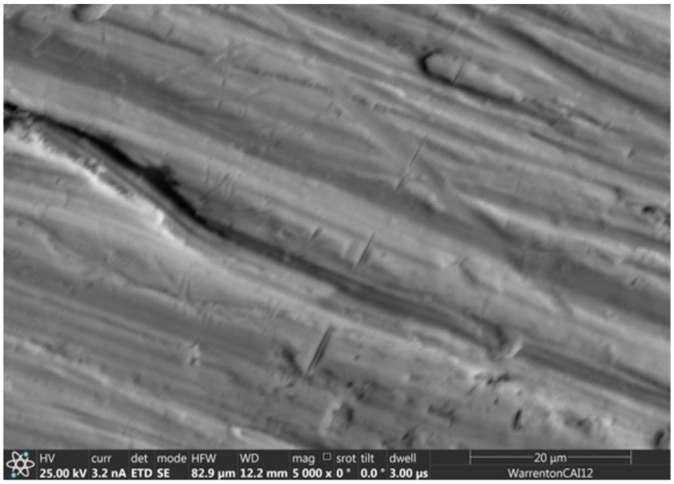


(a) As-Received

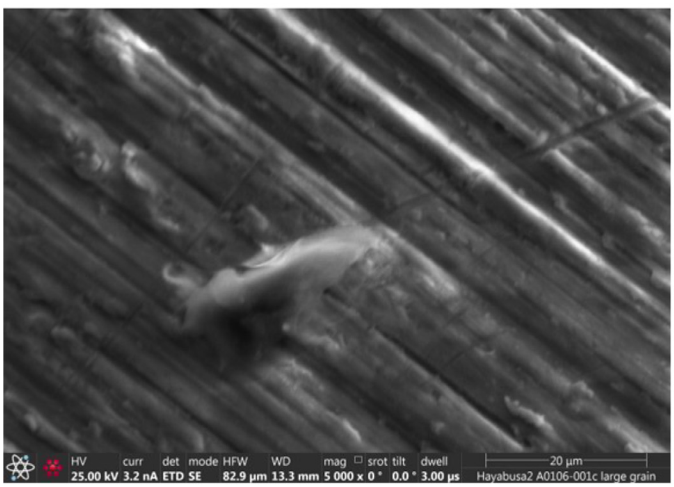


(b) Post-CHF

Fig. C11. SEM scan of Zircaloy 4 before and after pool boiling CHF experiment.



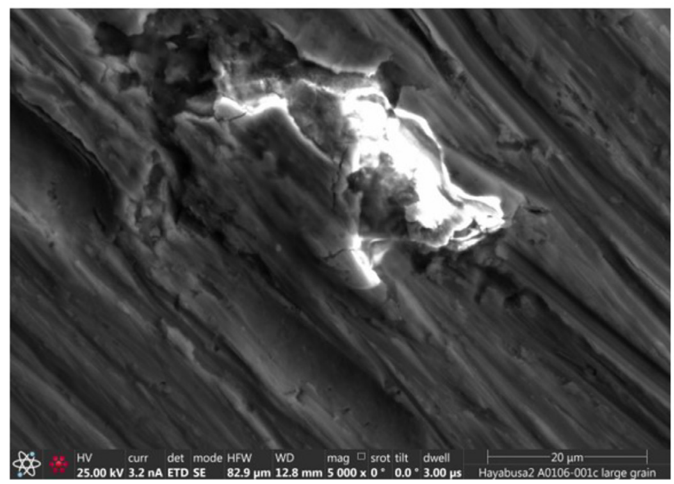
(a) As-Received



(b) Post-CHF

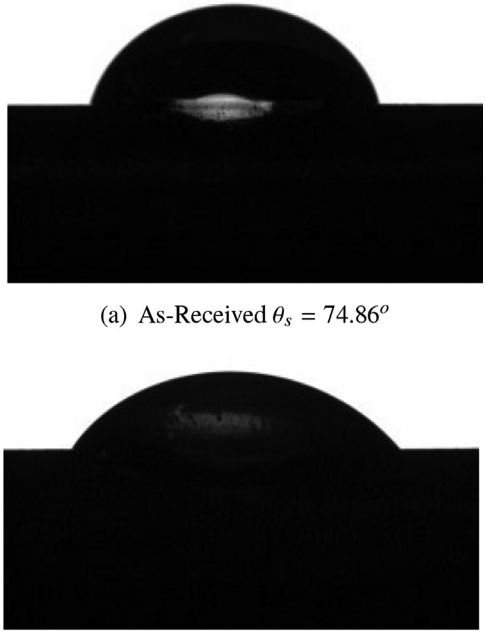
Fig. C12. SEM scan of Zirlo before and after pool boiling CHF experiment.

(a) As-Received



(b) Post-CHF

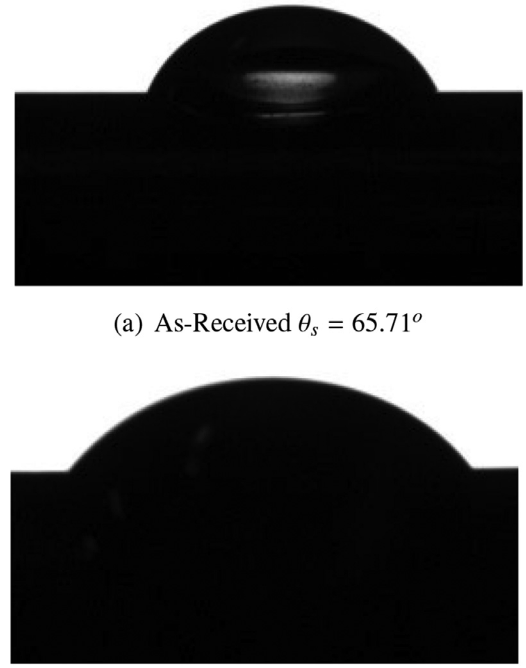
Fig. C13. SEM scan of Zr 705 before and after pool boiling CHF experiment.



(a) As-Received $\theta_s = 74.86^o$

(b) Post-CHF $\theta_s = 36.16^o$

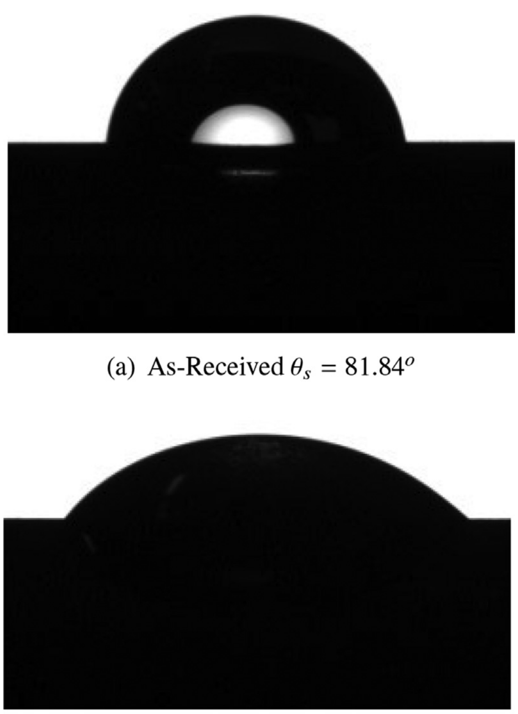
Fig. C14. Surface static contact angles of FeCrAl-B126Y before and after pool boiling CHF experiment.



(a) As-Received $\theta_s = 65.71^o$

(b) Post-CHF $\theta_s = 52.17^o$

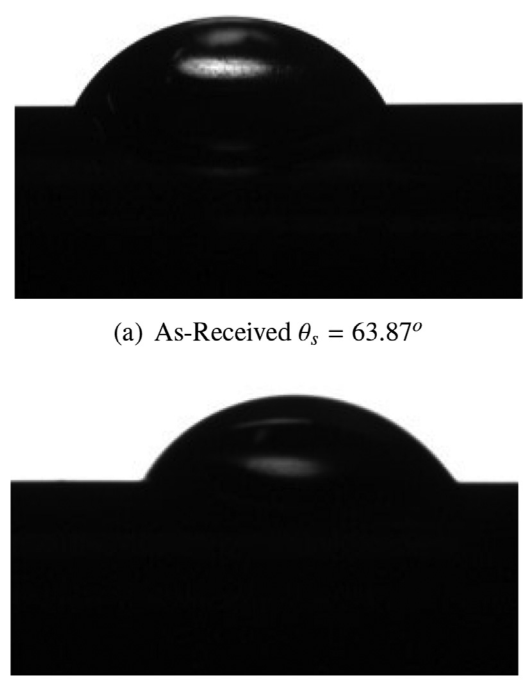
Fig. C16. Surface static contact angles of Zircaloy 4 before and after pool boiling CHF experiment.



(a) As-Received $\theta_s = 81.84^o$

(b) Post-CHF $\theta_s = 35.31^o$

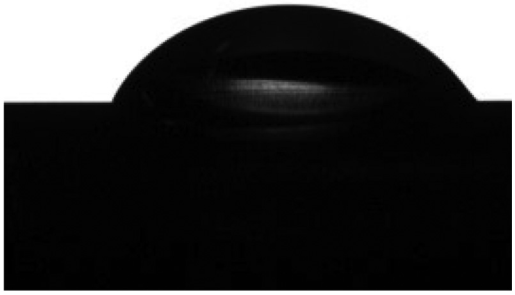
Fig. C15. Surface static contact angles of FeCrAl-B136Y before and after pool boiling CHF experiment.



(a) As-Received $\theta_s = 63.87^o$

(b) Post-CHF $\theta_s = 58.19^o$

Fig. C17. Surface static contact angles of Zirlo before and after pool boiling CHF experiment.



(a) As-Received $\theta_s = 57.11^o$

(b) Post-CHF $\theta_s = 54.62^o$

Fig. C18. Surface static contact angles of Zr 705 before and after pool boiling CHF experiment.

Appendix D. Boiling behaviors

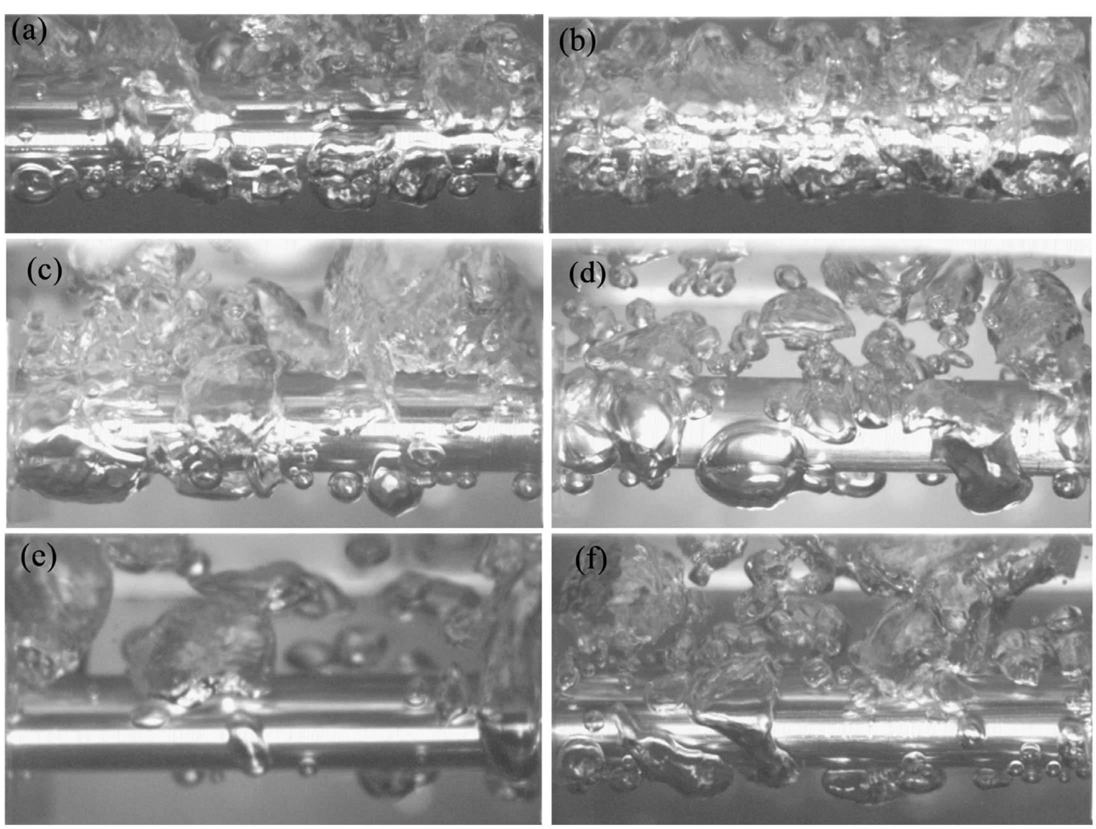


Fig. D1. Bubble behaviors of three Zircaloys and three FeCrAl alloys at the heat flux of 139 kW/m^2:(a) FeCrAl-B126Y, (b) FeCrAl-B136Y, (c), FeCrAl-C26M, (d) Zircaloy 4, (e) Zirlo, and (f) Zr 705.

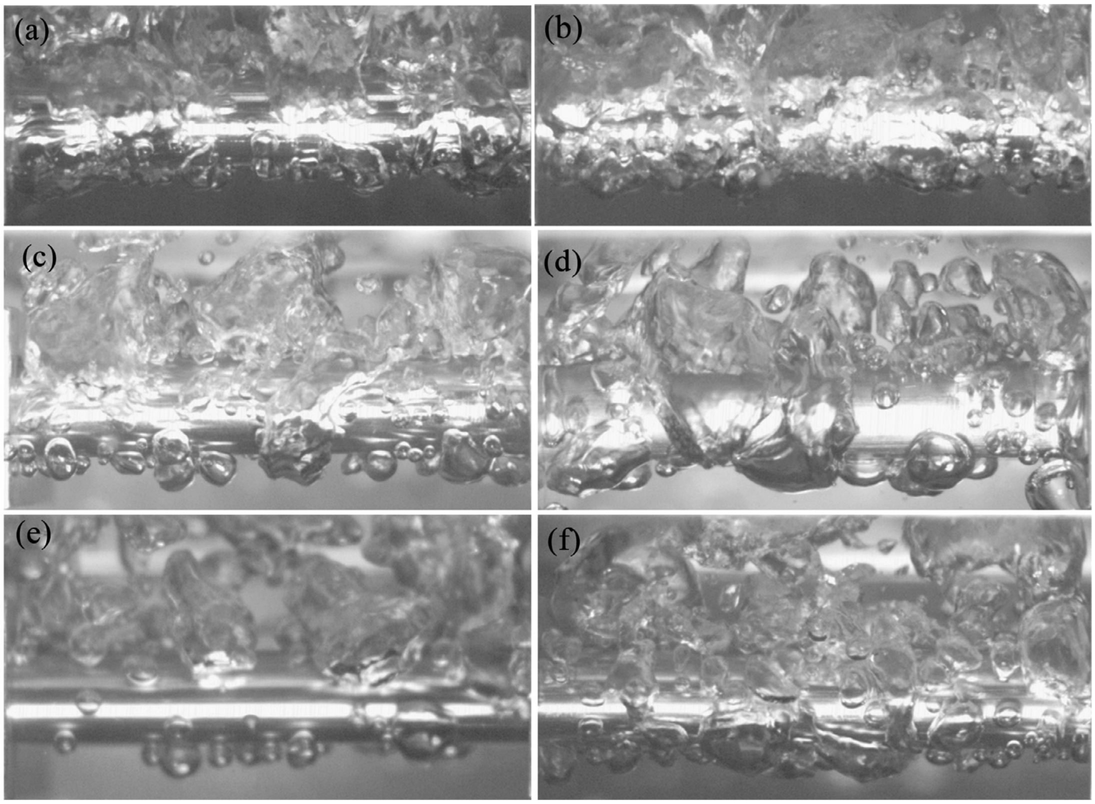


Fig. D2. Bubble behaviors of three Zircaloys and three FeCrAl alloys at the heat flux of 206 kW/m^2:(a) FeCrAl-B126Y, (b) FeCrAl-B136Y, (c), FeCrAl-C26M, (d) Zircaloy 4, (e) Zirlo, and (f) Zr 705.

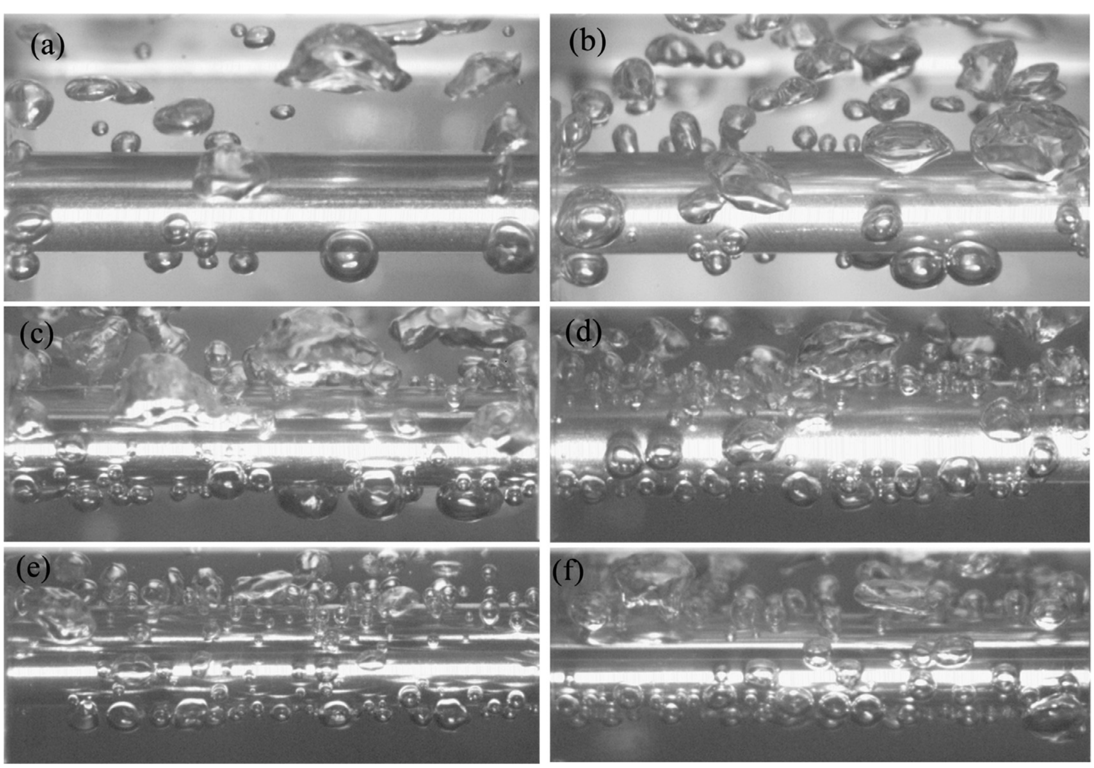


Fig. D3. Bubble behaviors of four stainless steels and two Inconels at the heat flux of 51 kW/m^2 (OD = 3/8″, WT = 0.02″): (a) Inconel 600, (b) Inconel 625, (c) SS 304, (d) SS 316, (e) SS 321, and (f) SS 347.

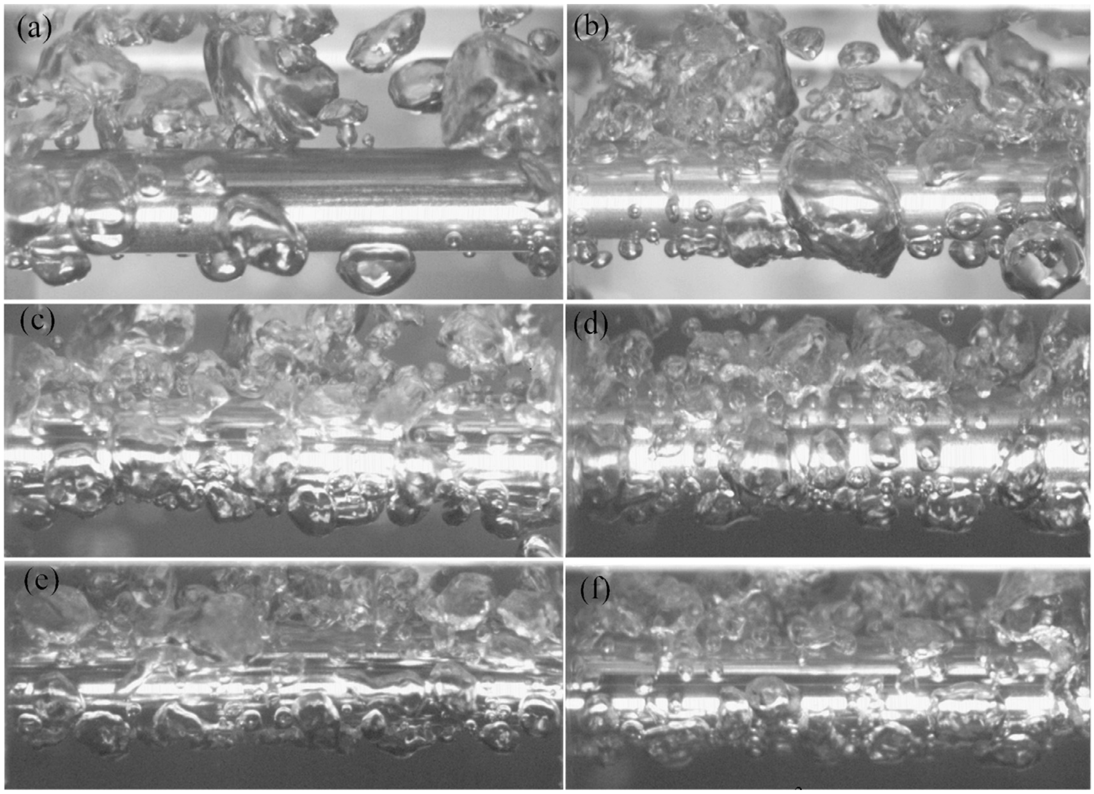


Fig. D4. Bubble behaviors of four stainless steels and two Inconels at the heat flux of 139 kW/m^2 (OD = 3/8″, WT = 0.02″): (a) Inconel 600, (b) Inconel 625, (c) SS 304, (d) SS 316, (e) SS 321, and (f) SS 347.

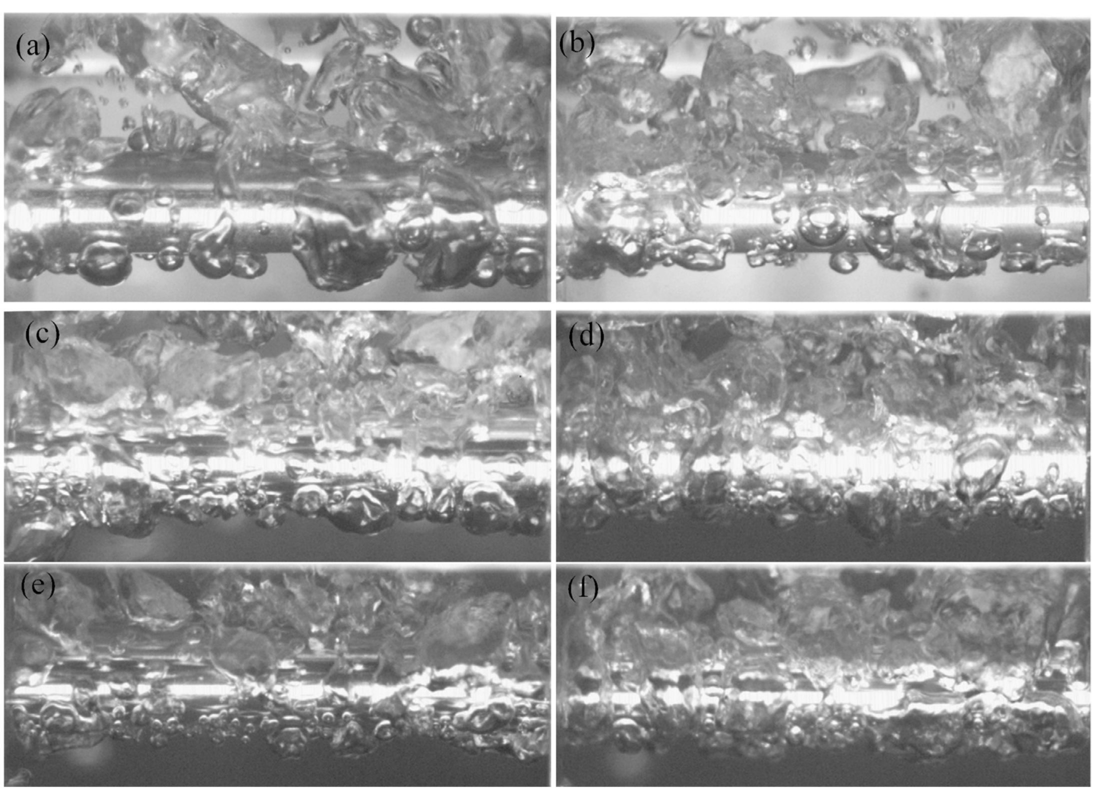


Fig. D5. Bubble behaviors of four stainless steels and two Inconels at the heat flux of 206 kW/m^2 (OD = 3/8″, WT = 0.02″): (a) Inconel 600, (b) Inconel 625, (c) SS 304, (d) SS 316, (e) SS 321, and (f) SS 347*.

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